UNIT - 1

SIMPLE STRESSES & STRAINS

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- Elasticity & plasticity, types of stresses & strains, Hooke's law
- Stress-strain diagram for mild steel, working stress, factor of safety
- Lateral strain, Poisson's ratio, volumetric strain
- Bars of varying section, composite bars, temperature stresses
- Complex stresses: inclined planes, principal planes & stresses, Mohr's circle
- Relation between elastic constants
- Strain energy, resilience: gradual, sudden, impact & shock loading

Tip: learn the formulas in the yellow boxes - most exam problems are direct substitution!

1. Introduction

Mechanics of solids (strength of materials) studies how a deformable body responds to loads - how much it deforms and whether it fails.

When a body is loaded, it develops an internal resisting force. This resisting force per unit area is called stress.

- Rigid body: shape does not change (ideal).
- Deformable body: changes shape and size under load (real).

- Aim of design: strength, stiffness and stability at minimum cost.

2. Elasticity and plasticity

Elasticity: the property by which a body regains its original size and shape when the load is removed.

Plasticity: the property by which a body retains permanent deformation after the load is removed.

- Perfectly elastic body – returns completely, e.g. steel within elastic limit, rubber.
- Plastic body – no recovery at all, e.g. wet clay, lead.
- Elastic limit: the maximum stress up to which the material is elastic.
- Ductile: large plastic deformation before fracture (mild steel, copper). Brittle: fractures with little strain (cast iron, glass, concrete).

3. Stress and its types

Stress σ = Load P / Area A unit: N/mm^2 = MPa

Type	Action of load	Formula
Tensile	pulls, bar elongates	$\sigma_t = P/A$
Compressive	pushes, bar shortens	$\sigma_c = P/A$
Shear	force parallel to the area	$\tau = P/A$
Bearing	contact pressure (rivets)	$\sigma_b = P/(d \cdot t)$

1 MPa = 1 N/mm^2 = 10^6 N/m^2; 1 GPa = 10^3 N/mm^2 = 10^9 Pa. E for steel = 2×10^5 N/mm^2.

4. Strain and its types

Strain is the change in dimension per unit original dimension. It has no units.

- Linear (tensile / compressive) strain $\varepsilon = \delta L / L$
- Shear strain $\varphi = \delta / h$ (angle of distortion in radians)
- Volumetric strain $\varepsilon_v = \delta V / V$

5. Hooke's law

Within the proportional limit, stress is directly proportional to the strain it produces.

$$\sigma \propto \varepsilon \quad \Rightarrow \quad \sigma = E\,\varepsilon \qquad E = \text{Young's modulus}$$
$$\tau = G\,\varphi \qquad K = \sigma / \varepsilon_v \text{ (bulk modulus)}$$

Combining $\sigma = P/A$ and $\varepsilon = \delta L/L$, the elongation of an axially loaded bar is:

$$\delta L = P L / (A E)$$

E, G and K are the three elastic moduli. A stiff material has a large E (steel 200 GPa, aluminium 70 GPa).

6. Stress-strain diagram for mild steel



Fig: Stress-strain curve of mild steel (schematic, not to scale)

- O-A: straight line, Hooke's law valid. A = proportional limit.
- A-B: elastic but not proportional. B = elastic limit.
- C, D: yielding. Upper yield point C, then stress drops to lower yield point D; strain increases with almost constant stress.

- D-E: strain hardening – the material stiffens, stress rises to the ultimate stress at E.
- E-F: necking – the area reduces locally and the bar breaks at F.

For materials with no clear yield point (aluminium, high-carbon steel) the 0.2 % proof stress (offset method) is used as yield stress.

> % elongation = $(L_f - L) / L \times 100$ % reduction in area = $(A - A_f) / A \times 100$

7. Working stress and factor of safety

The working (permissible) stress is the maximum stress allowed in service. It is kept below the failure stress to give a margin of safety.

> Factor of safety (FOS) = Ultimate stress / Working stress (for ductile materials, yield stress is used in place of ultimate stress)

Typical FOS: steel 3 to 4, cast iron 5 to 6, timber 7 to 10, dynamic loads larger.

Margin of safety = FOS – 1. Reasons for FOS: uncertain loads, material defects, workmanship, wear.

8. Lateral strain and Poisson's ratio

When a bar is stretched axially it also contracts laterally (and vice versa on compression).

> Poisson's ratio μ = lateral strain / longitudinal strain

- $\mu \approx 0.25$ to 0.33 for most metals (steel 0.3), concrete ≈ 0.15, rubber ≈ 0.5.
- Lateral strain is opposite in sign to the longitudinal strain: $\varepsilon_{lat} = -\mu \varepsilon$.

9. Volumetric strain

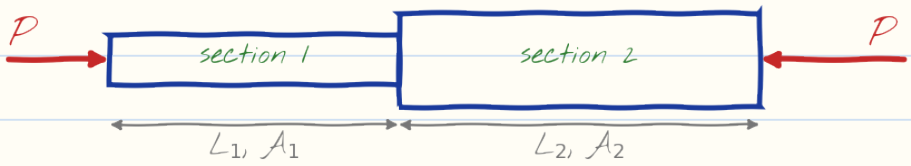
For a rectangular bar under axial stress σ along x:

$$\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z = \varepsilon\,(1 - 2\mu) \quad [\varepsilon = \sigma/E]$$

For a circular bar: $\varepsilon_v = \varepsilon_L + 2\,\varepsilon_d = \delta L/L + 2\,\delta d/d$. Since ε_v cannot be negative for tension, $\mu \le 0.5$.

10. Bars of varying section

A bar made of several lengths with different areas under the same axial load P (same material):



$$\delta L = \Sigma\, P\, L_i / (A_i\, E) = (P/E)\,[L_1/A_1 + L_2/A_2 + \ldots]$$

If loads act at the junctions, draw the axial-force diagram first and use the force of each segment.

- Tapered circular bar (d_1 to d_2): $\delta L = 4\,P\,L / (\pi\,E\,d_1\,d_2)$
- Tapered rectangular bar (constant thickness t, widths b_1, b_2): $\delta L = P\,L\,\ln(b_1/b_2) / [E\,t\,(b_1 - b_2)]$
- Bar hanging under own weight W: $\delta L = W\,L / (2\,A\,E) = \rho\,g\,L^2 / (2E)$

Example 1: stepped bar

A steel bar carries $P = 50$ KN. Section 1: $d = 20$ mm, $L = 500$ mm. Section 2: $d = 30$ mm, $L = 400$ mm. $E = 2\times10^5$ MPa. Find δL.

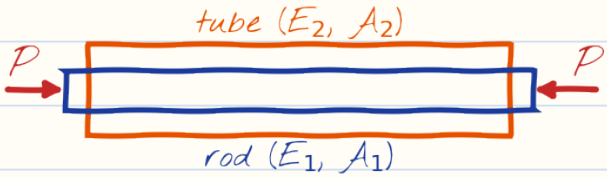
$A_1 = \pi/4 \times 20^2 = 314.16$ mm^2, $A_2 = \pi/4 \times 30^2 = 706.86$ mm^2

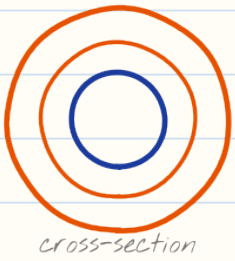
$\delta L = (50000 / 2\times10^5)\,[500/314.16 + 400/706.86]$

$= 0.25 \times [1.5915 + 0.5659] = 0.539$ mm

11. Composite bars

Two (or more) materials joined so that they extend or shorten together under a common load.





- Equilibrium: $P = P_1 + P_2 = \sigma_1 A_1 + \sigma_2 A_2$
- Compatibility: $\varepsilon_1 = \varepsilon_2 \Rightarrow \sigma_1/E_1 = \sigma_2/E_2$ (equal change in length)

$$P_1 = P \times A_1E_1 / (A_1E_1 + A_2E_2) \qquad \sigma_1/\sigma_2 = E_1/E_2 = \text{modular ratio } m$$

Example 2: steel rod in copper tube

Steel rod $d = 25$ mm inside a copper tube (40 mm outer, 30 mm inner) carries $P = 100$ KN compression. $E_s = 200$ GPa, $E_c = 100$ GPa.

$A_s = 490.87\ mm^2$, $A_c = \pi/4\,(40^2 - 30^2) = 549.78\ mm^2$

$\sigma_s = 2\,\sigma_c$ and $100000 = \sigma_c\,(2 \times 490.87 + 549.78) = 1531.5\,\sigma_c$

$\sigma_c = 65.3$ MPa, $\sigma_s = 130.6$ MPa

Check: $P_s = 64.1$ KN, $P_c = 35.9$ KN, sum = 100 KN ✓

12. Temperature stresses

A bar free to expand grows by $\delta = \alpha T L$ (no stress). If the expansion is prevented, a thermal stress develops:

$$\sigma = \alpha T E \qquad P = \alpha T E A$$

(α = coefficient of thermal expansion, T = temperature rise)

- Rise in temperature + fixed ends → compressive stress; fall in temperature → tensile stress.
- If the support yields by δ: $\sigma = E(\alpha T L - \delta)/L$.

- Composite bar heated: the material with larger α is put in compression, the other in tension.

$$\sigma_1 A_1 = \sigma_2 A_2 \text{ (equilibrium)} \qquad \sigma_1/E_1 + \sigma_2/E_2 = (\alpha_2 - \alpha_1) T \text{ (compatibility)}$$

Example 3

A steel rod of 25 mm dia is fixed between rigid walls at 20 °C and heated to 70 °C. $\alpha = 12\times10^{-6}$ /°C, $E = 200$ GPa.

$\sigma = \alpha T E = 12\times10^{-6} \times 50 \times 2\times10^{5} = 120$ MPa (compressive)

$P = \sigma A = 120 \times 490.87 = 58\,904\ N \approx 58.9$ KN

13. Complex stresses – stresses on an inclined plane

Case 1: uniaxial stress σ. On a plane whose normal makes angle θ with the axis:

$$\sigma_n = \sigma \cos^2\theta \qquad \tau_t = (\sigma/2) \sin 2\theta$$

Maximum shear stress = $\sigma/2$, occurring on a plane at 45° to the axis. Max normal stress = σ at $\theta = 0$.

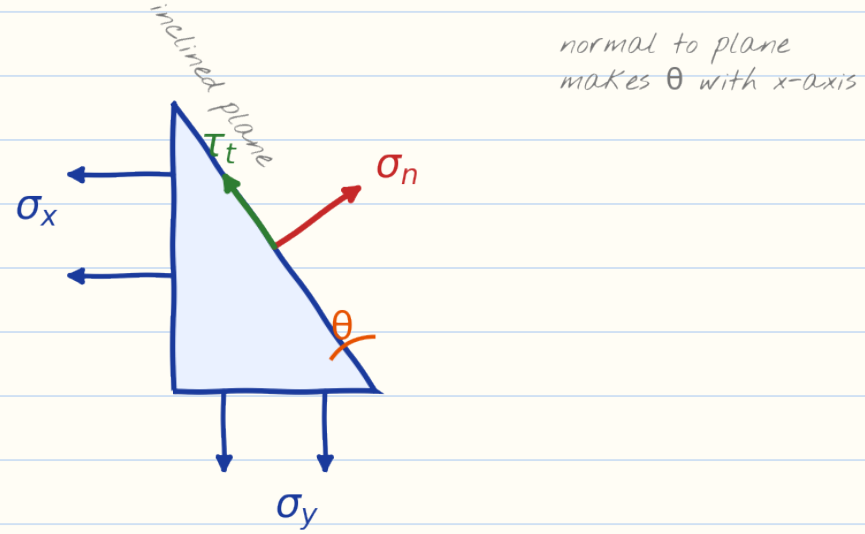


Fig: element cut by an inclined plane (normal at to x-axis)

Case 2: biaxial stresses σ_x, σ_y

$$\sigma_n = (\sigma_x+\sigma_y)/2 + (\sigma_x-\sigma_y)/2 \cdot \cos 2\theta$$
$$\tau_t = (\sigma_x-\sigma_y)/2 \cdot \sin 2\theta$$

Case 3: biaxial + shear τ_{xy}

$$\sigma_n = (\sigma_x+\sigma_y)/2 + (\sigma_x-\sigma_y)/2 \cdot \cos 2\theta + \tau_{xy} \sin 2\theta$$
$$\tau_t = (\sigma_x-\sigma_y)/2 \cdot \sin 2\theta - \tau_{xy} \cos 2\theta$$

- Resultant stress on the plane $\sigma_R = \sqrt{(\sigma_n^2 + \tau_t^2)}$
- Obliquity (angle of resultant with the normal): $\tan \varphi = \tau_t / \sigma_n$

14. Principal planes and principal stresses

Principal planes are planes on which the shear stress is zero. The normal stresses on them are the principal stresses.

$$\tan 2\theta_p = 2\tau_{xy} / (\sigma_x - \sigma_y)$$
$$\sigma_{1,2} = (\sigma_x+\sigma_y)/2 \pm \sqrt{[((\sigma_x-\sigma_y)/2)^2 + \tau_{xy}^2]}$$
$$\tau_{max} = \sqrt{[((\sigma_x-\sigma_y)/2)^2 + \tau_{xy}^2]} = (\sigma_1 - \sigma_2)/2$$

- The two principal planes are 90° apart, the planes of max shear are at 45° to them.
- On planes of max shear the normal stress is $(\sigma_x+\sigma_y)/2$.

Example 4

σ_x = 100 MPa, σ_y = 40 MPa (both tensile), τ_{xy} = 30 MPa. Find principal stresses and the stresses on a plane at θ = 30°.

average = 70, $R = \sqrt{(30^2 + 30^2)} = 42.43$

σ_1 = 112.43 MPa, σ_2 = 27.57 MPa, τ_{max} = 42.43 MPa

$\tan 2\theta_p = 60/60 = 1 \Rightarrow 2\theta_p = 45°$, $\theta_p = 22.5°$

at 30°: $\sigma_n = 70 + 30 \cos 60° + 30 \sin 60° = 110.98$ MPa

$\tau_t = 30 \sin 60° - 30 \cos 60° = 10.98$ MPa

15. Mohr's circle

A graphical method - every point on the circle represents the (σ, τ) on some plane. Angles on the circle are double the real angles.

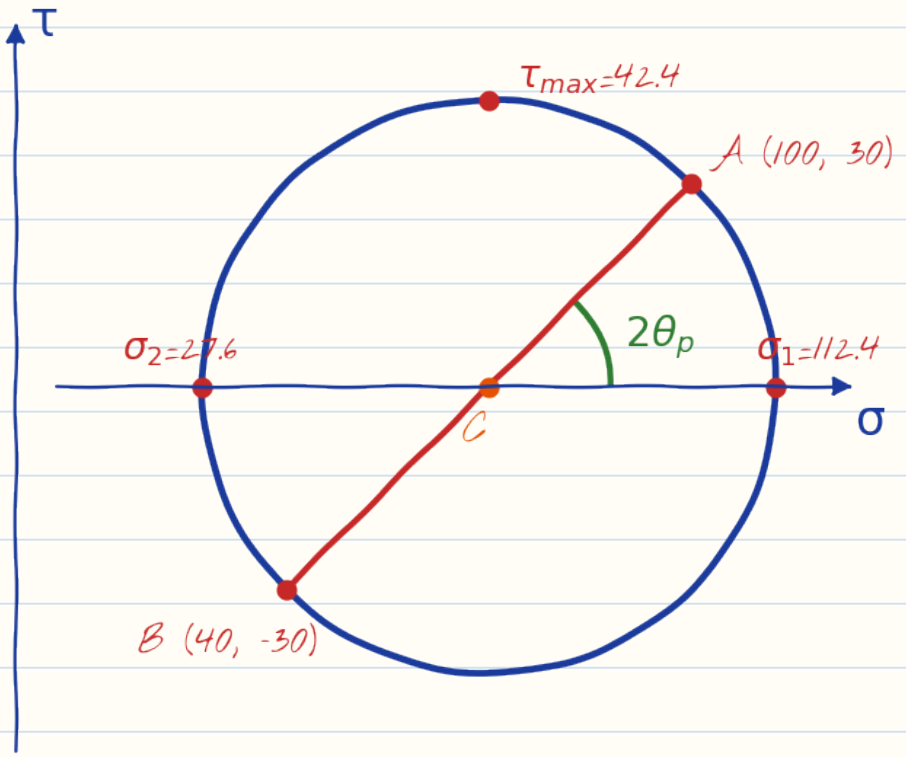


Fig: Mohr's circle for Example 4

- Step 1: choose scale, plot A (σ_x, $+\tau_{xy}$) and B (σ_y, $-\tau_{xy}$).
- Step 2: join AB, it cuts the σ-axis at the centre C = $(\sigma_x+\sigma_y)/2$.
- Step 3: draw the circle with diameter AB. Radius = τ_{max}.
- Step 4: the points where the circle cuts the σ-axis give σ_1 and σ_2.
- Step 5: for a plane at θ, rotate CA by 2θ (same sense), the coordinates of that point are σ_n, τ_t.

16. Relation between elastic constants

$E = 2G(1+\mu)$	$E = 3K(1-2\mu)$
$E = 9KG/(3K+G)$	$\mu = (3K-2G)/(6K+2G)$

Proof of $E = 2G(1+\mu)$: in pure shear τ the principal stresses are $+\tau$ and $-\tau$ at 45°. Strain along $+\tau$ direction: $\varepsilon = \tau/E + \mu\tau/E = (\tau/E)(1+\mu)$. This equals half the shear strain: $\varphi/2 = \tau/2G$. Hence $E = 2G(1+\mu)$.

Proof of $E = 3K(1-2\mu)$: under equal stress σ on all faces $\varepsilon_v = 3\sigma/E(1-2\mu)$ and $\varepsilon_v = \sigma/K$. Hence $E = 3K(1-2\mu)$.

Example 5

$E = 2\times10^5$ MPa, $G = 0.8\times10^5$ MPa. Find μ and K.

$\mu = E/2G - 1 = 2/1.6 - 1 = 0.25$

$K = E / 3(1 - 2\mu) = 2\times10^5 / 1.5 = 1.333\times10^5$ MPa

17. Strain energy and resilience

Work stored in a deformed elastic body = strain energy U. For gradual loading, $U = \frac{1}{2} P \delta L$ (area under the load-extension line).

$$U = \sigma^2 V / (2E) \qquad [V = A L = \text{volume}]$$

- Proof resilience: maximum strain energy stored (at elastic limit).
- Modulus of resilience = $\sigma_p^2 / 2E$ (energy per unit volume).
- Shear strain energy: $U = \tau^2 V / (2G)$.

18. Gradual, sudden, impact and shock loading

Loading	Stress	Elongation
Gradual (static)	$\sigma = P/A$	$\delta = PL/AE$
Sudden (suddenly applied)	$\sigma = 2P/A$	$\delta = 2PL/AE$
Impact (fall through h)	$\sigma = (P/A)[1 + \sqrt{(1 + 2hAE/PL)}]$	$\delta = \sigma L/E$

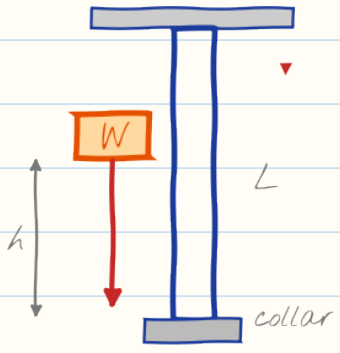


Fig: weight W falling through height h on a collar

- Sudden load produces twice the stress of the same load applied gradually.
- If h is large compared with δ: $\sigma \approx \sqrt{(2 W h E / A L)}$.
- Shock load: a sudden/impact load whose effect is judged by the energy absorbed.

Example 6: impact load

$W = 2$ KN falls $h = 30$ mm on a collar at the end of a 3 m long, 20 mm dia bar. $E = 200$ GPa.

$A = 314.16$ mm^2, $W/A = 6.366$ MPa, $\delta_{st} = WL/AE = 0.0955$ mm

$\sigma = 6.366\,[1 + \sqrt{(1 + 2 \times 30/0.0955)}] = 6.366 \times 26.09 = 166$ MPa

More solved problems

Example 7: factor of safety

A mild steel bar (yield stress 250 MPa) is to carry a tensile load with FOS = 2.5. Find the safe load for $d = 20$ mm.

$\sigma_w = 250 / 2.5 = 100$ MPa, $A = 314.16$ mm^2

$P = \sigma_w A = 100 \times 314.16 = 31.4$ KN

Example 8: Poisson's ratio and volume change

A 20 mm dia, 2 m long bar carries 40 KN tension. $E = 2 \times 10^5$ MPa, $\mu = 0.3$. Find δL, δd and δV.

$\sigma = 40000/314.16 = 127.3$ MPa, $\varepsilon = \sigma/E = 6.366 \times 10^{-4}$

$\delta L = \varepsilon L = 6.366 \times 10^{-4} \times 2000 = 1.273$ mm

$\delta d = -\mu \varepsilon d = -0.3 \times 6.366 \times 10^{-4} \times 20 = -0.0038$ mm

$\varepsilon_v = \varepsilon (1 - 2\mu) = 6.366 \times 10^{-4} \times 0.4 = 2.546 \times 10^{-4}$

$V = 314.16 \times 2000 = 6.283 \times 10^5$ mm^3 $\Rightarrow$ $\delta V = 160$ mm^3

Example 9: composite bar heated

Steel rod ($A = 600$ mm^2, $E = 200$ GPa, $\alpha = 12 \times 10^{-6}$) and copper rod ($A = 900$ mm^2, $E = 100$ GPa, $\alpha = 17 \times 10^{-6}$) are rigidly joined at both ends and heated by 60 °C. Find stresses.

Equilibrium: $\sigma_s \times 600 = \sigma_c \times 900$ $\Rightarrow$ $\sigma_s = 1.5\,\sigma_c$

Compatibility: $\sigma_s/E_s + \sigma_c/E_c = (\alpha_c - \alpha_s)\,T = 5\times10^{-6} \times 60 = 3\times10^{-4}$

$1.5\sigma_c/2\times10^5 + \sigma_c/10^5 = 3\times10^{-4} \Rightarrow 3.5\,\sigma_c = 60$

$\sigma_c = 17.14$ MPa (compressive), $\sigma_s = 25.71$ MPa (tensile)

Check forces: $25.71 \times 600 = 17.14 \times 900 = 15.43$ KN ✓

Example 10: inclined plane, uniaxial

A bar carries a uniaxial tensile stress of 100 MPa. Find σ_n and τ on a plane whose normal is at 30° to the axis.

$\sigma_n = 100\cos^2 30° = 75$ MPa

$\tau = (100/2)\sin 60° = 43.3$ MPa (maximum shear = 50 MPa at 45°)

Example 11: strain energy

A bar 1 m long, $A = 500$ mm², carries 50 KN gradually. $E = 200$ GPa. Find strain energy stored.

$\sigma = 100$ MPa, $V = 500 \times 1000 = 5\times10^5$ mm³

$U = \sigma^2 V / 2E = 100^2 \times 5\times10^5 / (2 \times 2\times10^5) = 12\,500$ N·mm = 12.5 N·m (J)

If the same load is applied suddenly: $\sigma = 200$ MPa, $U = 4 \times 12.5 = 50$ J.

Important questions (university pattern)

- Draw the stress-strain curve for mild steel and explain salient points.
- Define Poisson's ratio and derive the relation between E, G and K.
- Derive the expression for stress on an inclined plane under biaxial stress with shear.
- Explain Mohr's circle and use it to find principal stresses for a given state of stress.
- Derive the elongation of a bar of uniformly tapering circular section.

- A composite bar is heated - derive stresses in the two materials.
- Derive the expression for stress due to a falling weight (impact load).

Quick revision - Unit I

$\sigma = P/A$	$\varepsilon = \delta L/L$	$\delta L = PL/AE$	$FOS = \sigma_{"}/\sigma_{""}$	μ = lateral/long.
$\sigma_{temp} = \alpha TE$	$\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[\ldots]}$	$U = \sigma^2 V/2E$		

- Important: stress-strain curve of mild steel; relation between E, G, K; Mohr's circle problems.
- Important: composite bar & temperature stress problems; principal stress problems; impact loading.

UNIT - II

SHEAR FORCE & BENDING MOMENT

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- Definition of a beam, types of beams, supports and loads
- Concept of shear force (S.F.) and bending moment (B.M.), sign conventions
- S.F. and B.M. diagrams: cantilever, simply supported and overhanging beams
- Point loads, u.d.l., uniformly varying load (u.v.l.) and combinations
- Point of contra-flexure
- Relation between S.F., B.M. and rate of loading

Tip: SFD and BMD questions carry good marks. Learn the standard results and the step-by-step method.

1. Beam - definition

A beam is a structural member, usually horizontal, whose length is large compared with its cross-section and which is loaded transverse (perpendicular) to its axis. Its main action is bending.

- Loads on beams cause internal shear force and bending moment at every section.
- Beam analysis: find reactions → find S.F. and B.M. → draw diagrams → design.

2. Types of beams

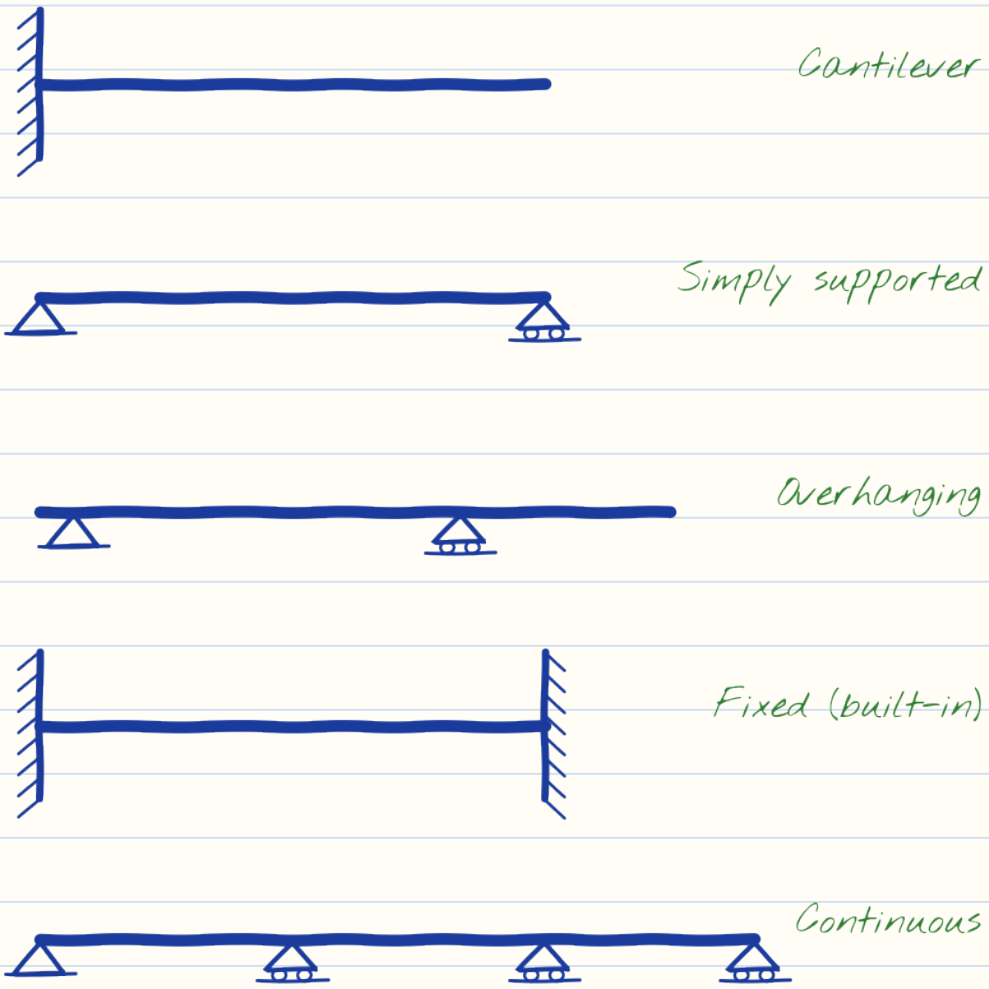


Fig: Common types of beams

- Cantilever: one end fixed, the other end free.
- Simply supported: rests on a hinge at one end and a roller at the other.
- Overhanging: a simply supported beam extended beyond one or both supports.
- Fixed (built-in): both ends fixed - restrained against rotation.
- Continuous: supported at more than two points.

Statically determinate beams: reactions found from equilibrium alone (cantilever, simply supported, overhanging). Fixed and continuous beams are statically indeterminate.

3. Types of supports

Support	Reactions	Restrains
Roller	one (vertical)	vertical movement
Hinge (pin)	two (V and H)	vertical + horizontal
Fixed	V, H and moment	movement + rotation

4. Types of loads

- Point (concentrated) load W - acts at a point, unit KN.
- Uniformly distributed load (u.d.l.) w - constant intensity, KN/m. Equivalent point load = w x length, acting at the centre of the load.
- Uniformly varying load (u.v.l.) - intensity varies linearly. Triangular load: total = ½ x base x peak, acts at 1/3 from the larger end.
- Couple (moment) applied at a point, unit KN·m.

5. Shear force and bending moment

Shear force at a section = the algebraic sum of all vertical forces acting on either side of the section.

Bending moment at a section = the algebraic sum of the moments of all forces on either side of the section about that section.

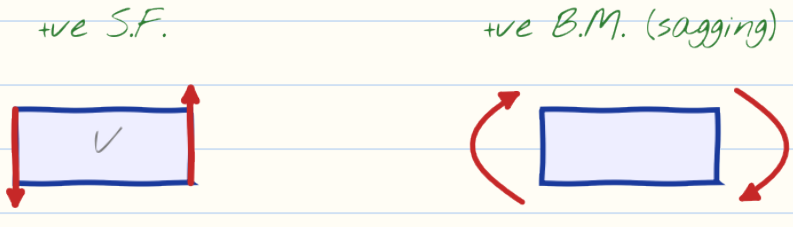


Fig: Sign conventions

- S.F. positive: left part tends to move up relative to right (clockwise shear). Forces on the left upward, or on the right downward, give +ve S.F.

- B.M. positive (sagging): bends the beam concave upward - tension at bottom. Negative = hogging, tension at the top.

Quick rule (working from the left): upward force → +ve S.F.; downward force → -ve S.F. Moments of upward forces to the left give +ve (sagging) B.M.

6. Relations between w, S.F. and B.M.

Consider a small element dx carrying load w per unit length:

$$\frac{dV}{dx} = -w \qquad \frac{dM}{dx} = V \qquad \frac{d^2M}{dx^2} = -w$$

- The slope of the S.F. diagram = (−) load intensity.
- The slope of the B.M. diagram = S.F. at that section.
- B.M. is maximum (or minimum) where S.F. = 0 or changes sign.
- Change in S.F. between two sections = area of the load diagram between them.
- Change in B.M. between two sections = area of the S.F. diagram between them.
- At a point load, S.F. jumps by the load; at an applied couple, B.M. jumps by the couple.

Load on span	S.F.D.	B.M.D.
No load	horizontal line	inclined straight line
Point load	sudden jump	change of slope (kink)
u.d.l.	inclined straight line	parabola
u.v.l.	parabola	cubic curve

7. Point of contra-flexure

The point of contra-flexure (point of inflexion) is the section where the B.M. changes sign - i.e. where B.M. = 0 and the curvature of the beam reverses (sagging to hogging).

- Found by equating the B.M. expression to zero.

- Exists mostly in overhanging and fixed/continuous beams. Cantilevers and simply supported beams with downward loads have none.

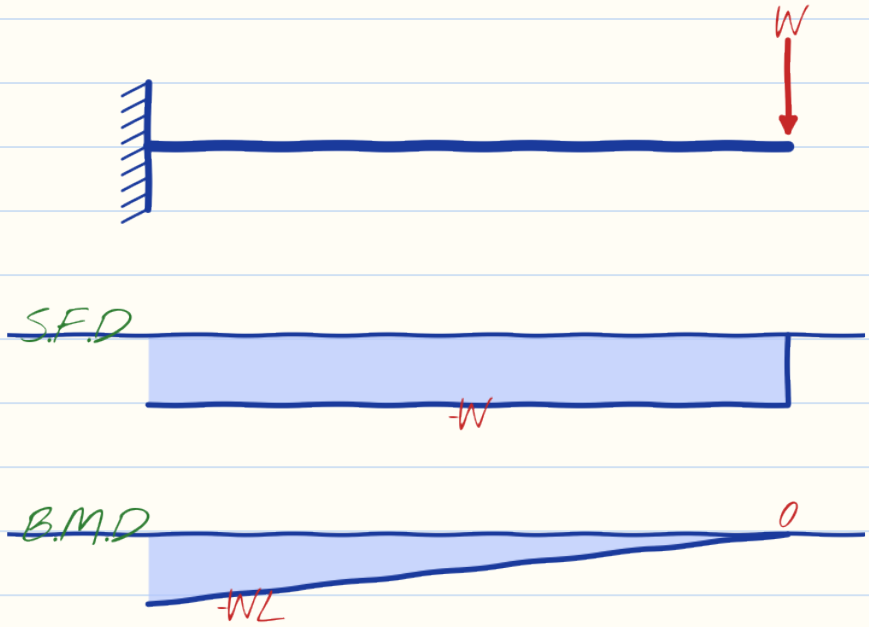
8. Procedure for drawing S.F.D. and B.M.D.

- 1. Replace distributed loads by equivalent point loads to find reactions ($\Sigma V = 0$, $\Sigma M = 0$).
- 2. Divide the beam into portions at every point load, support, and start/end of u.d.l.
- 3. Write V and M for each portion using one side of the section.
- 4. Calculate V and M at the ends of each portion and plot.
- 5. Join with straight line / curve as per the load. Put sign, value and units.
- 6. Locate the section of zero shear for the maximum bending moment.

9. Cantilever beams

(a) Point load W at the free end

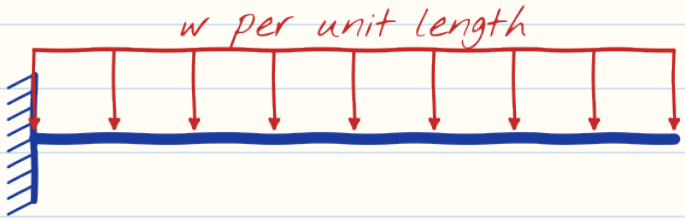
Take x from the free end: $V = -W$ (constant), $M = -Wx$.



$V_{max} = -W$ $M_{max} = -WL$ (at the fixed end)

(b) u.d.l. w over the whole span

$V = -wx$ (straight line), $M = -wx^2/2$ (parabola).







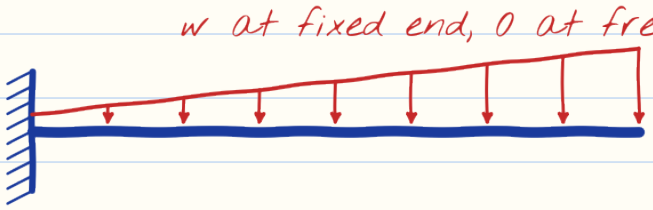
$V_{max} = -wL$ $M_{max} = -wL^2/2$

(c) u.v.l. – zero at the free end, w per metre at the fixed end

Intensity at x from free end = wx/L. Load up to $x = \frac{1}{2}(wx/L)x$.

$V = -wx^2/2L$ $M = -wx^3/6L$

at fixed end: $V = -wL/2$, $M = -wL^2/6$







Example 1: cantilever with combined loads

A cantilever of length 4 m carries a point load of 10 KN at the free end and a u.d.l. of 5 KN/m over the whole length. Find V and M at the fixed end.

V at free end = − 10 KN

V at fixed end = − (10 + 5×4) = − 30 KN

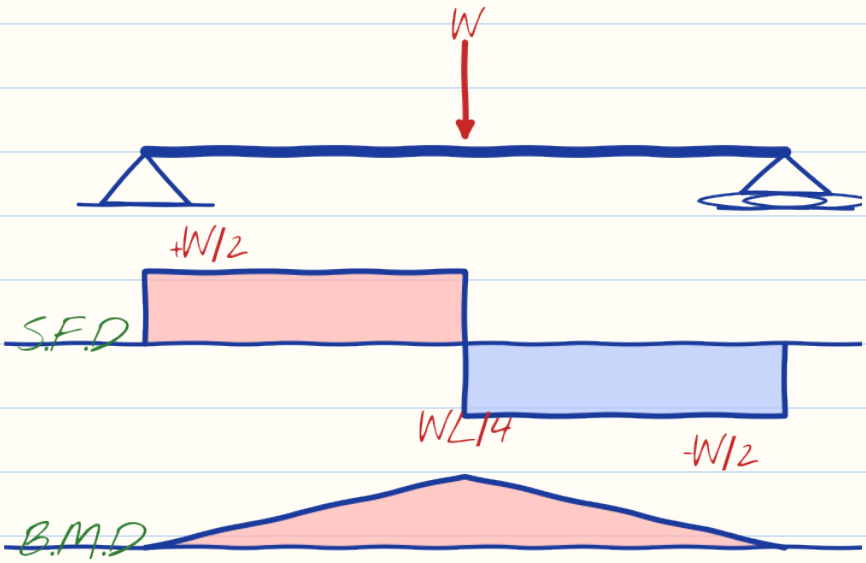
M at fixed end = − (10×4 + 5×4×2) = − (40 + 40) = − 80 KN·m

Both diagrams lie below the base line (hogging): V is straight (u.d.l.), M is a parabola.

10. Simply supported beams

(a) Central point load W

$R_A = R_B = W/2$. For $x < L/2$: $V = W/2$, $M = W x/2$. For $x > L/2$: $V = -W/2$.



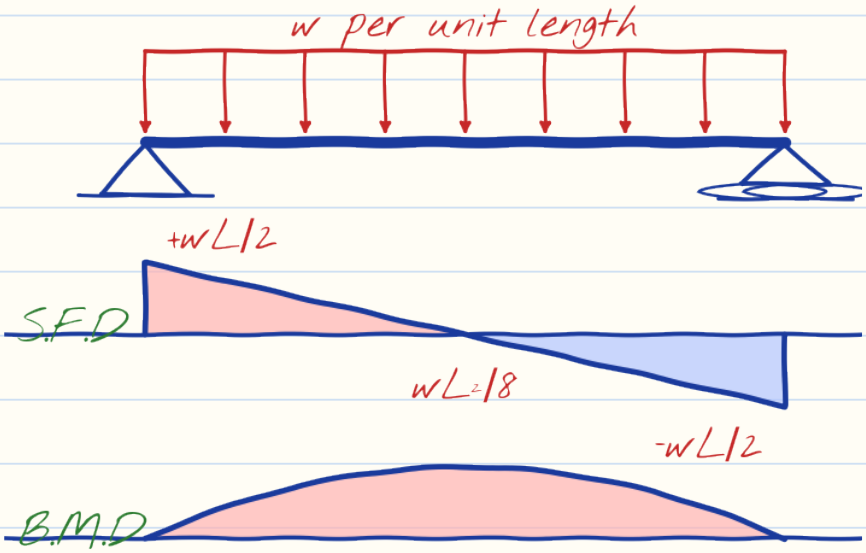
$$M_{max} = W L / 4 \text{ at the centre}$$

For an eccentric load W at distance a from A and b from B ($L = a + b$):

$$R_A = W b / L \qquad R_B = W a / L$$
$$M_{max} = W a b / L \text{ (under the load)}$$

(b) u.d.l. w over the full span

$R_A = R_B = wL/2$. $V = wL/2 - w\,x$ (zero at $x = L/2$). $M = wL\,x/2 - w\,x^2/2$.



$$M_{max} = w\,L^2 / 8 \text{ at the centre}$$

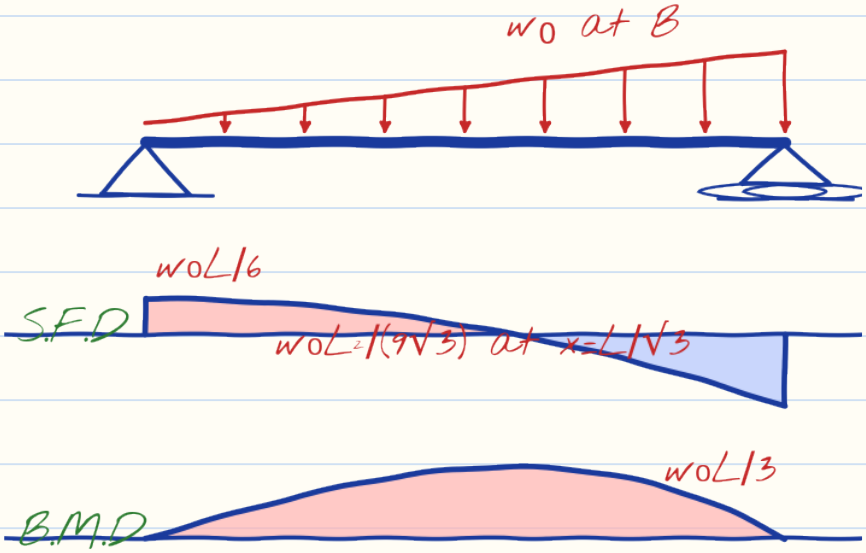
(c) u.v.l. – zero at A, w_0 per metre at B

Total load = $w_0\,L/2$ acting at $L/3$ from the heavy end B.

$$R_A = w_0L/6 \qquad R_B = w_0L/3$$

$V = w_0L/6 - w_0x^2/2L$ (zero at $x = L/\sqrt{3} = 0.577\,L$)

$M_{max} = w_0L^2/(9\sqrt{3}) = 0.0642\,w_0L^2$



Example 2: simply supported beam with point load and u.d.l.

Beam AB = 8 m. Point load 20 KN at 2 m from A; u.d.l. 5 KN/m from 4 m to 8 m (over 4 m). Draw SFD and BMD.

Moments about A: $R_B \times 8 = 20 \times 2 + (5 \times 4) \times 6 = 160 \Rightarrow R_B = 20$ KN

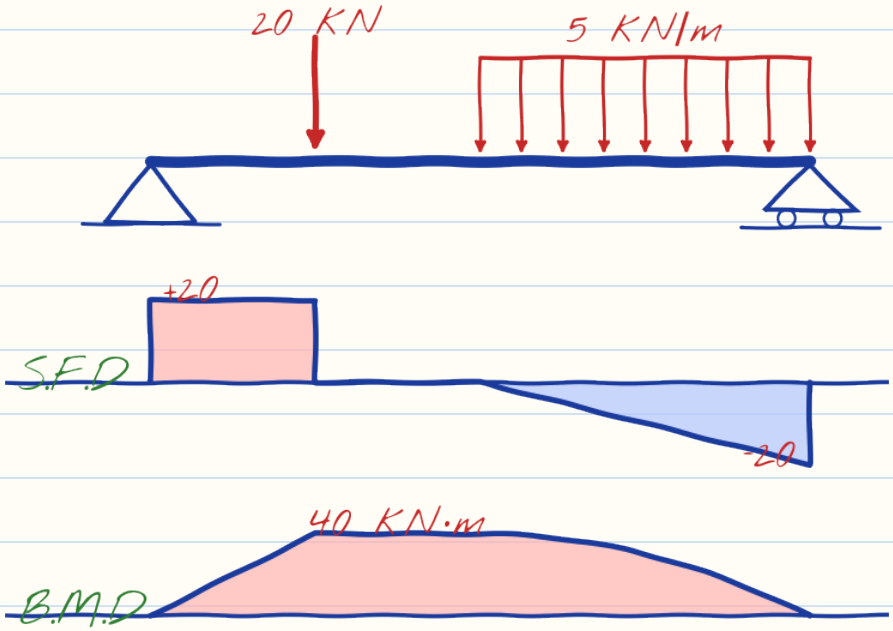
$R_A = 20 + 20 - 20 = 20$ KN

S.F.: A: +20; just left of C (2 m): +20; just right: 0; constant 0 up to 4 m;

then falls at 5 KN/m to $0 - 5 \times 4 = -20$ KN at B.

B.M.: at 2 m = $20 \times 2 = 40$; at 4 m = 40 (S.F. is zero between 2 and 4 m);

at 8 m = $40 - 5 \times 4^2/2 = 0$ ✓. $M_{max} = 40$ KN·m between C and D.



II. Overhanging beams

An overhanging beam has a part projecting beyond a support. The overhang gives a negative (hogging) B.M. over the support and hence a point of contra-flexure in the span.

Example 3: overhanging beam with u.d.l.

A beam 8 m long rests on supports A and B 6 m apart (overhang BC = 2 m) and carries a u.d.l. of 10 KN/m over the whole 8 m. Draw SFD and BMD, locate the point of contra-flexure.

Total load = $10 \times 8 = 80$ KN at 4 m from A.

$\Sigma M_A = 0$: $R_B \times 6 = 80 \times 4 \Rightarrow R_B = 53.33$ KN, $R_A = 26.67$ KN

S.F.: A = +26.67; just left of B = 26.67 − 60 = −33.33;

just right of B = −33.33 + 53.33 = +20; at C = 20 − 20 = 0 ✓

B.M. in span: $M = 26.67x - 5x^2$ (x from A)

V = 0 at $x = 26.67/10 = 2.667$ m: $M_{max} = 35.56$ KN·m

M at B = 26.67×6 − 5×36 = −20 KN·m (hogging) = − 10×2×1 ✓

Contra-flexure: $26.67x - 5x^2 = 0 \Rightarrow x = 5.33$ m from A

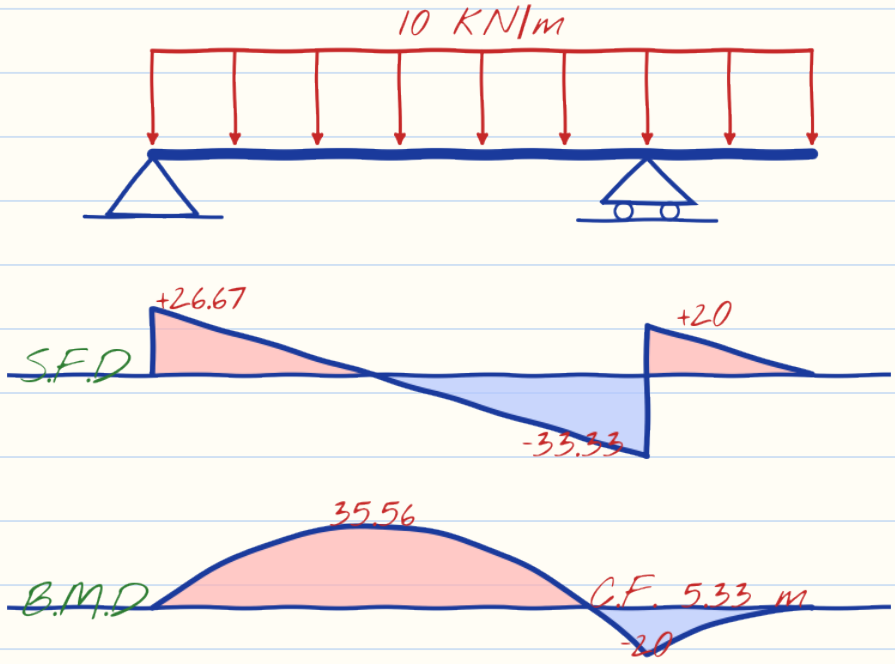


Fig. S.F.D. and B.M.D. of Example 3

12. Effect of an applied couple

- A couple M_0 does not change the S.F. at its point.

- The B.M. diagram jumps by M_0 at the point: a clockwise couple on the left part gives an upward jump (sagging +ve).
- Reactions of a simply supported beam with a couple M_0: $R_A = -M_0/L$, $R_B = +M_0/L$ (clockwise couple).

More solved problems

Example 4: S.S. beam, two point loads

Span AB = 6 m; 10 KN at 2 m and 20 KN at 4 m from A.

$R_B \times 6 = 10\times2 + 20\times4 = 100 \Rightarrow R_B = 16.67$ KN, $R_A = 13.33$ KN

S.F.: 0–2 m: +13.33; 2–4 m: 13.33 − 10 = +3.33; 4–6 m: 3.33 − 20 = −16.67

B.M.: at 2 m = 13.33×2 = 26.67; at 4 m = 26.67 + 3.33×2 = 33.33; at B = 33.33 − 16.67×2 = 0 ✓

$M_{max} = 33.33$ KN·m under the 20 KN load (S.F. changes sign there).

Example 5: cantilever with partial u.d.l.

Cantilever 3 m: u.d.l. 4 KN/m over the 2 m next to the fixed end, point load 6 KN at the free end.

Free end to 1 m: V = −6 (constant), then V decreases linearly to $-6 - 4\times2 = -14$ KN at the wall.

M at start of u.d.l. (1 m from free end) = $-6 \times 1 = -6$ KN·m

M at wall = $-(6\times3 + 8\times1) = -26$ KN·m (u.d.l. resultant 8 KN acts 1 m from the wall)

Example 6: S.S. beam with triangular load

Span 6 m, load increases from zero at A to 12 KN/m at B.

Total W = ½ × 6 × 12 = 36 KN, acting 2 m from B (4 m from A).

$R_A = 36 \times 2/6 = 12$ KN, $R_B = 24$ KN

$V = 12 - 12x^2/(2\times6) = 12 - x^2$; $V = 0$ at $x = \sqrt{12} = 3.464$ m

$M_{max} = 12 \times 3.464 - 3.464^3/3 = 41.57 - 13.86 = 27.71$ KN·m (= $w_0L^2/9\sqrt{3}$ ✓)

Tips and common mistakes

- Always check: reactions sum to the total load and $\Sigma M = 0$ about the other support.
- S.F. diagram must close to zero at the free end / right support. B.M. must be zero at simple supports and free ends.
- Distributed loads: convert to a point load only for finding reactions or for a section beyond the load.
- For a section inside a u.d.l., use only the part of the load to the left (or right) of the section.
- Area method: M at a point = M at the previous point + area of S.F. diagram between them.
- Do not forget the units - KN, KN/m, KN·m.

13. Shape of the diagrams - summary

Beam & load	V_{max}	M_{max}
Cantilever, W at end	W	W L
Cantilever, u.d.l.	w L	$w L^2 / 2$
Cantilever, u.v.l. (0 at free end)	$w L / 2$	$w L^2 / 6$
S.S., central W	$W / 2$	$W L / 4$
S.S., u.d.l.	$w L / 2$	$w L^2 / 8$
S.S., u.v.l. (0 at A)	$w L / 3$	$0.0642\, w L^2$

Practice problems (answers given)

- S.S. beam, span 10 m, u.d.l. 2 KN/m over the whole span. Find reactions and M_{max}. Ans: 10 KN each, 25 KN·m
- Cantilever 2 m with a 5 KN point load at the free end. Find M at the fixed end. Ans: -10 KN·m
- S.S. beam, span 4 m, central point load 8 KN. Find reactions and M_{max}. Ans: 4 KN each, 8 KN·m

- Cantilever 3 m with u.d.l. 6 KN/m. Find V and M at the fixed end. Ans: -18 KN, -27 KN·m
- S.S. beam, span 6 m, 24 KN at 2 m from A. Find R_A, R_B, M_{max}. Ans: 16 KN, 8 KN, 32 KN·m

Important questions

- Define S.F. and B.M. State the sign conventions.
- Derive the relation between load intensity, shear force and bending moment.
- Draw SFD and BMD for a simply supported beam with u.d.l. and show $M_{max} = wL^2/8$.
- What is a point of contra-flexure? Locate it for an overhanging beam.
- Draw the SFD and BMD for a cantilever loaded with a point load, u.d.l. and u.v.l.

UNIT - III

FLEXURAL & SHEAR STRESSES

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- FLEXURAL STRESSES: theory of simple bending, derivation of the bending equation
- Bending stresses, section modulus of rectangular, circular, I and T sections
- Design of simple beam sections
- SHEAR STRESSES: derivation of the shear stress formula
- Shear stress distribution across rectangular, circular, triangular, I and T sections

Tip: $M/I = \sigma/y = E/R$ and $\tau = V A \bar{y} / (I b)$ are the two master equations of this unit.

1. Theory of simple bending

When a beam carries transverse loads, the bending moment makes it curve. The top fibres shorten (compression) and the bottom fibres lengthen (tension) for a sagging moment. Between them lies a layer that neither shortens nor lengthens - the *neutral surface*. Its trace on the cross-section is the *neutral axis* (N.A.).

Pure bending: a beam portion subjected to constant B.M. and zero shear force (e.g. the middle third of a beam loaded by two equal loads symmetric about the centre).

2. Assumptions in the theory of simple bending

- The material is homogeneous and isotropic.
- Hooke's law holds - stresses are within the elastic limit.
- E is the same in tension and in compression.
- Transverse sections plane before bending remain plane after bending.
- The beam is initially straight and has a constant cross-section, the radius of curvature is large compared with the depth.
- Each layer is free to expand or contract independently of the layers above and below.
- The beam is bent in a plane of symmetry, loads act in that plane.

3. Derivation of the bending equation

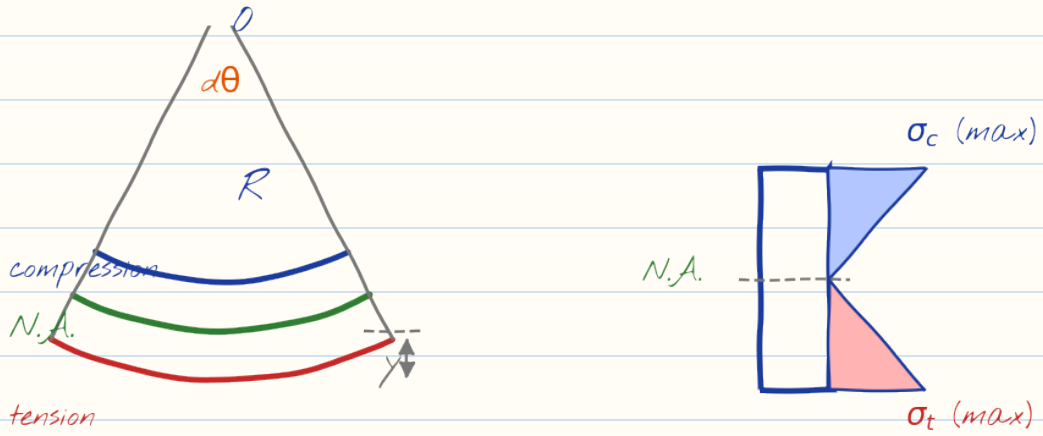


Fig: bending of a beam element, stress distribution on the section

Consider a small length of beam bent to a circular arc of radius R (to the neutral axis), subtending angle dθ at the centre O.

Length of N.A. layer = R dθ (unchanged)

Length of a layer at distance y below N.A. = (R + y) dθ

Strain $\varepsilon = [(R + y)\,d\theta - R\,d\theta] / R\,d\theta = y / R$

Stress $\sigma = E\,\varepsilon = E\,y / R$ i.e. $\sigma / y = E / R$ ...(1)

Stress varies linearly with y: zero at N.A., maximum at the extreme fibre.

Position of N.A.: no net axial force on the section, so

$$\int \sigma \, dA = (E/R) \int y \, dA = 0 \quad \Rightarrow \quad \int y \, dA = 0$$

i.e. the neutral axis passes through the centroid of the section.

Moment of resistance: moment of the force $\sigma \, dA$ about N.A. is $\sigma y \, dA$.

$$M = \int \sigma y \, dA = (E/R) \int y^2 \, dA = E \, I / R \qquad \ldots(2)$$

$$M / I = \sigma / y = E / R$$

M = bending moment (N·mm), I = M.I. of the section about N.A. (mm^4), σ = bending stress (N/mm^2), y = distance from N.A. (mm), R = radius of curvature (mm).

4. Section modulus

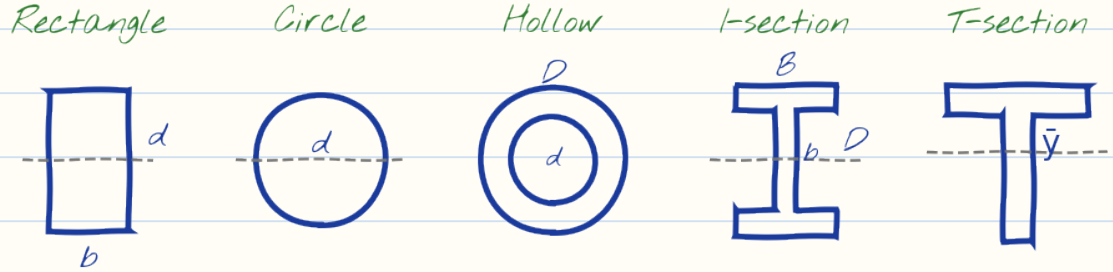
The maximum stress occurs at the extreme fibre, $y = y_{max}$:

$$\sigma_{max} = M \, y_{max} / I = M / Z \qquad Z = I / y_{max}$$

(section modulus, mm^3)

- Z measures the strength of a section in bending: $M_r = \sigma_{allow} Z$.
- For an unsymmetrical section (T, L) there are two moduli: $Z_{top} = I/y_t$ and $Z_{bottom} = I/y_b$.
- A section with more material far from N.A. (I-section) has a larger Z for the same area.

5. M.I. and section modulus of standard sections



Section	I about N.A.	$Z = I / y_{max}$
Rectangle $b \times d$	$b\,d^3 / 12$	$b\,d^2 / 6$
Solid circle, dia d	$\pi\,d^4 / 64$	$\pi\,d^3 / 32$
Hollow circle D, d	$\pi\,(D^4 - d^4) / 64$	$\pi\,(D^4 - d^4) / 32D$
Triangle b, h (about centroid)	$b\,h^3 / 36$	$b\,h^2 / 24$ (at base)
Symmetrical I-section	$(B\,D^3 - b\,d^3) / 12$	$I / (D/2)$

For I-section B = flange width, D = overall depth, b = (B – web thickness) and d = clear depth between flanges if you use $I = (B\,D^3 - b\,d^3)/12$ (the hollow-out method). For built-up sections use the parallel axis theorem $I = I_G + A\,h^2$.

6. Worked examples

Example 1: rectangular beam

A simply supported timber beam 100 mm × 200 mm, span 4 m, carries a u.d.l. w. Permissible bending stress = 8 MPa. Find w.

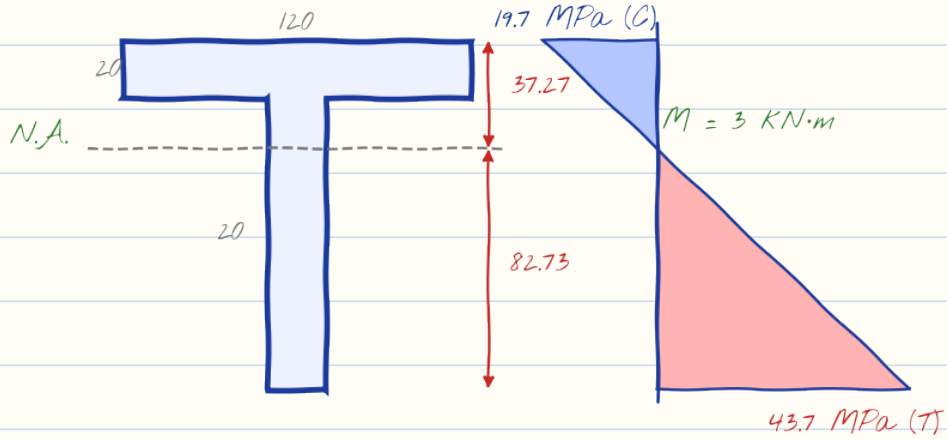
$Z = b\,d^2 / 6 = 100 \times 200^2 / 6 = 6.667 \times 10^5\ mm^3$

$M_{max} = \sigma\,Z = 8 \times 6.667 \times 10^5 = 5.333 \times 10^6\ N\cdot mm = 5.333\ KN\cdot m$

$w\,L^2/8 = 5.333 \Rightarrow w = 8 \times 5.333 / 4^2 = 2.67\ KN/m$

Example 2: T-section

T-section: flange 120 mm × 20 mm (top), web 20 mm × 100 mm. Find the bending stresses for a sagging moment of 3 KN·m.



$A_1 = 2400$ ($y = 10$ from top), $A_2 = 2000$ ($y = 70$)

$\bar{y} = (2400 \times 10 + 2000 \times 70) / 4400 = 37.27$ mm from top

$I = [120 \times 20^3/12 + 2400 \times 27.27^2] + [20 \times 100^3/12 + 2000 \times 32.73^2]$

$= 80\,000 + 1\,784\,880 + 1\,666\,667 + 2\,142\,400 = 5.674 \times 10^6$ mm^4

$\sigma_{top} = 3 \times 10^6 \times 37.27 / 5.674 \times 10^6 = 19.7$ MPa (compressive)

$\sigma_{bottom} = 3 \times 10^6 \times 82.73 / 5.674 \times 10^6 = 43.7$ MPa (tensile)

7. Design of simple beam sections

Design steps:

- 1. Find M_{max} from the B.M.D. (and include self-weight if required).
- 2. Required section modulus $Z_{req} = M_{max} / \sigma_{allow}$.
- 3. Choose the section so that $Z_{provided} \geq Z_{req}$.
- 4. Check shear stress, deflection and local buckling.

Rectangular ($d = 2b$): $b = (3Z / 2)^{1/3}$ Circular: $d = (32Z / \pi)^{1/3}$

Example 3: design

Design a rectangular ($d = 2b$) and a circular section for $M_{max} = 30$ KN·m and $\sigma_{allow} = 150$ MPa.

$Z_{req} = 30 \times 10^6 / 150 = 2 \times 10^5$ mm^3

Rectangular: $b = (1.5 \times 2 \times 10^5)^{1/3} = 66.9$ mm $\Rightarrow$ 70 mm x 140 mm ($A = 9800$ mm^2)

Circular: $d = (32 \times 2 \times 10^5 / \pi)^{1/3} = 126.8$ mm $\Rightarrow$ $d = 130$ mm ($A = 13\,273$ mm^2)

For the same strength the circular section needs about 35 % more material. An I-section is more economical still - most material is placed far from the N.A.

8. Shear stresses in beams – derivation

A transverse shear force V causes shear stress τ on the cross-section. By complementary shear the same τ acts on horizontal planes. Consider a layer at distance y from the N.A.

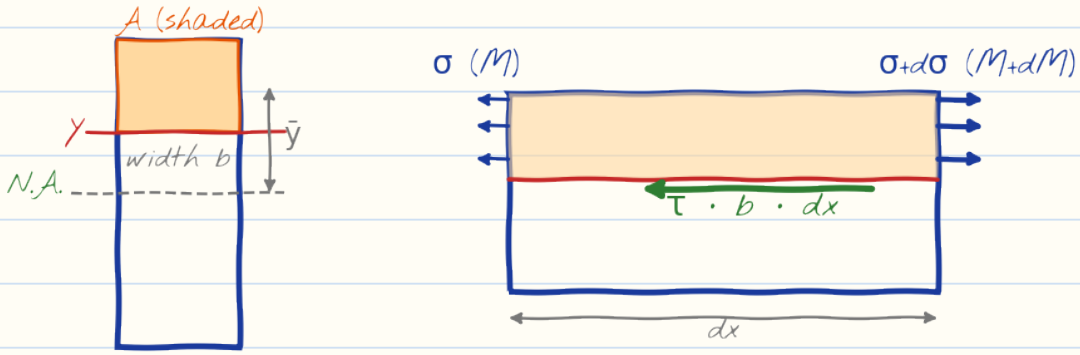


Fig: element of length dx, shaded area A above the level y

Force on the left face of the shaded part = $\int \sigma \, dA = (M/I) \int y \, dA$

Force on the right face = $((M + dM)/I) \int y \, dA$

Difference = $(dM/I) \int y \, dA = (dM/I) A \bar{y}$ (resisted by shear on the bottom face)

$\tau \, b \, dx = (dM/I) A \bar{y}$ and $dM/dx = V$

$$\tau = V A \bar{y} / (I b)$$

- V = S.F. at the section, A = area of the section <u>above</u> (or below) the level considered,
- $\bar{y}$ = distance of the centroid of A from the N.A., I = M.I. of whole section about N.A.,
- b = width of the section at the level considered.
- τ is maximum at the N.A. in most sections and zero at the extreme fibres.

9. Shear stress distribution

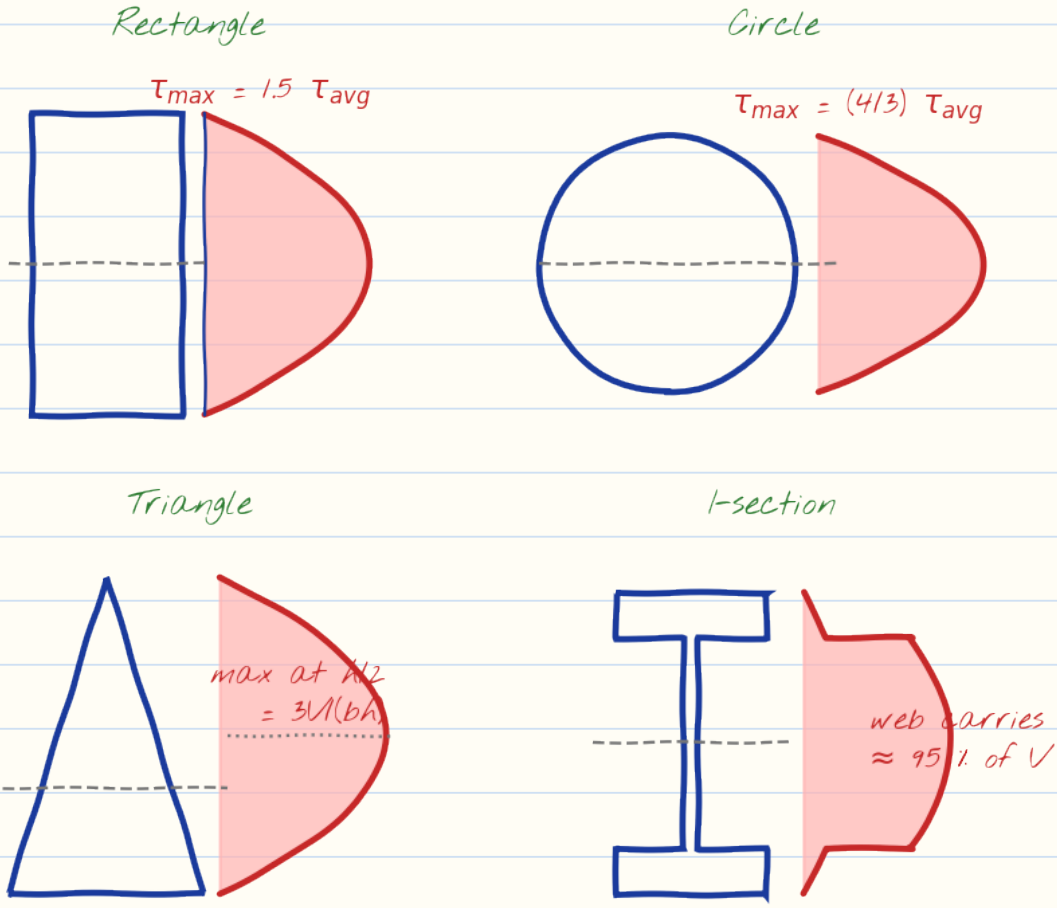


Fig: Shear stress distributions

(a) Rectangular section b x d

$\tau = (V/2I)(d^2/4 - y^2)$ - parabolic.

$\tau_{max} = 3V / 2A = 1.5\ \tau_{avg}$ (at N.A.) $\tau = 0$ at $y = \pm d/2$

(b) Circular section, diameter d

$\tau = (4V / 3\pi R^2)(1 - y^2/R^2)$ - parabolic.

$\tau_{max} = 4V / 3A = 1.333\ \tau_{avg}$ (at N.A.)

(c) Triangular section (base b, height h, apex up)

$\tau = 12\ V y (h - y) / (b\ h^3)$ (y measured from apex).

$\tau_{max} = 3V / (b\ h) = 1.5\ \tau_{avg}$ at h/2 from apex; at N.A. (2h/3): $\tau = 8V / 3bh = 1.33\ \tau_{avg}$

The maximum shear stress in a triangle is NOT at the neutral axis - it occurs at mid-height.

(d) I-section (flange B, depth D, web b, clear web depth d)

- In the web the variation is parabolic, in the flanges it is small (flange is wide).
- At the web-flange junction there is a sudden jump (the width changes from B to b).
- τ at N.A. (max) = $V / (8 / b) [B (D^2 - d^2) + b d^2]$
- τ at junction (in web) = $V B (D^2 - d^2) / (8 / b)$, inside the flange = (b/B) of that value.
- The web carries about 95 % of the shear force; flanges carry most of the bending moment.

(e) T-section

Find N.A. ($\bar{y}$), then use $\tau = V A \bar{y} / (I b)$ at the level required. Maximum τ at the N.A.; for the flange-web junction compute two values (with b = flange width, and b = web thickness) - a jump occurs.

10. Worked examples on shear

Example 4: rectangular beam

A rectangular beam 100 mm x 200 mm carries a shear force of 30 KN. Find τ_{avg} and τ_{max}.

$\tau_{avg} = 30000 / (100 \times 200) = 1.5$ MPa

$\tau_{max} = 1.5 \times 1.5 = 2.25$ MPa at the neutral axis

Example 5: I-section

I-section: flanges 100 mm x 20 mm, web 10 mm thick, overall depth 200 mm. V = 100 KN. Find shear stresses at N.A. and at the web-flange junction.

$I = (100 \times 200^3 - 90 \times 160^3)/12 = (8 \times 10^8 - 3.686 \times 10^8)/12 = 3.595 \times 10^7$ mm^4

At N.A.: $A \bar{y} = 100 \times 20 \times 90 + 10 \times 80 \times 40 = 212\,000$ mm^3

$\tau_{NA} = 100000 \times 212000 / (3.595 \times 10^7 \times 10) = 59.0$ MPa

At junction (web side): $A\bar{y} = 180\,000 \Rightarrow \tau = 100000 \times 180000 / (3.595 \times 10^8) = 50.1$ MPa

Just inside the flange: $\tau = 50.1 \times 10/100 = 5.0$ MPa (large jump!)

Average web shear = 100000 / (10 × 200) = 50 MPa – close to $\tau_{max} = 59$ MPa.

More solved problems

Example 6: bending stress in a circular beam

A simply supported beam, span 3 m, circular section d = 100 mm, central point load 10 KN. Find the maximum bending stress.

$M_{max} = W L / 4 = 10 \times 3 / 4 = 7.5$ KN·m $= 7.5 \times 10^6$ N·mm

$Z = \pi d^3 / 32 = \pi \times 100^3 / 32 = 98\,175$ mm^3

$\sigma_{max} = M / Z = 7.5 \times 10^6 / 98\,175 = 76.4$ MPa

Example 7: radius of curvature

A steel wire of 5 mm dia is bent round a drum of radius 2.5 m. $E = 2 \times 10^5$ MPa. Find the maximum stress in the wire.

$\sigma / y = E / R \Rightarrow \sigma = E y / R = 2 \times 10^5 \times 2.5 / 2500 = 200$ MPa

Here the radius is to the N.A. (use R + d/2 if the drum radius is to the inner surface).

Example 8: hollow circular section

Hollow shaft-type beam D = 100 mm, d = 80 mm carries M = 5 KN·m. Find σ_{max}.

$I = \pi (100^4 - 80^4) / 64 = \pi \times 5.904 \times 10^7 / 64 = 2.898 \times 10^6$ mm^4

$Z = I / 50 = 57\,962$ mm^3

$\sigma_{max} = 5 \times 10^6 / 57\,962 = 86.3$ MPa

Example 9: bending + shear check

S.S. beam, span 6 m, u.d.l. 20 KN/m, rectangular section 150 mm × 300 mm. Find σ_{max} and τ_{max}.

$M_{max} = w L^2 / 8 = 20 \times 36 / 8 = 90$ KN·m; $V_{max} = w L / 2 = 60$ KN

$Z = 150 \times 300^2 / 6 = 2.25 \times 10^6$ mm^3 $\Rightarrow$ $\sigma_{max} = 90 \times 10^6 / 2.25 \times 10^6 = 40$ MPa

$\tau_{max} = 1.5 \times 60000 / (150 \times 300) = 2.0$ MPa

Bending stress governs in slender beams; shear matters for short, deep beams and thin webs.

Example 10: circular section shear

A circular beam of dia 100 mm carries $V = 50$ KN. Find τ_{avg} and τ_{max}.

$A = \pi/4 \times 100^2 = 7854$ mm^2, $\tau_{avg} = 50000 / 7854 = 6.37$ MPa

$\tau_{max} = (4/3) \times 6.37 = 8.49$ MPa at the neutral axis

Comparison of sections

Section (same area)	Relative Z	Remark
Circle	1.00	least efficient
Square (sides vertical)	≈ 1.18	better than circle
Rectangle d = 2b	≈ 1.67	deep, narrow beams
I-section	3 to 5	most economical

- Flexural rigidity = E I. A deeper section increases I with the cube of depth.
- Rectangular beam: placing it with depth vertical (d > b) gives a much stronger beam than flat.
- Section modulus Z, not I alone, decides the moment a section can carry.

Example 11: shear in the T-section of Example 2

Same T-section ($I = 5.674 \times 10^6$ mm^4, N.A. 37.27 mm from top), $V = 20$ KN. Find τ at N.A. and at the flange-web junction.

Below N.A. (easier): web depth 82.73 mm, $A = 20 \times 82.73 = 1654.6$ mm^2, $\bar{y} = 41.37$ mm

$A\bar{y} = 68\,440\ mm^3 \Rightarrow \tau_{NA} = 20000 \times 68440 / (5.674\times10^6 \times 20) = 12.06\ MPa$

Junction: A = flange $2400\ mm^2$, $\bar{y} = 27.27\ mm$, $A\bar{y} = 65\,448\ mm^3$

in the web ($b = 20$): $\tau = 20000\times65448 / (5.674\times10^6\times20) = 11.53\ MPa$

in the flange ($b = 120$): $\tau = 11.53 \times 20/120 = 1.92\ MPa$

Practice problems (answers given)

- Rectangular beam 80 x 160 mm carries $M = 6\ KN\cdot m$. Find σ_{max}. Ans: $Z = 3.413\times10^5\ mm^3$, $\sigma = 17.6\ MPa$
- Solid circular beam $d = 60$ mm, permissible stress 100 MPa. Find the moment of resistance. Ans: 2.12 KN·m
- Rectangular beam 60 x 120 mm, $V = 18\ KN$. Find τ_{avg} and τ_{max}. Ans: 2.5 MPa, 3.75 MPa
- If $\sigma_{max} = 120\ MPa$ occurs at $y = 100$ mm, find σ at $y = 40$ mm. Ans: 48 MPa
- A steel strip 2 mm thick is bent to a radius of 1 m. Find the maximum stress ($E = 200\ GPa$). Ans: $\sigma = Ey/R = 2\times10^5 \times 1/1000 = 200\ MPa$

Important questions

- State the assumptions of simple bending and derive $M/I = \sigma/y = E/R$.
- Show that the neutral axis passes through the centroid of the section.
- Define section modulus, obtain it for rectangular, circular and hollow circular sections.
- Derive the shear stress formula and draw the distribution for a rectangular section.

- Draw the shear stress distribution for an I-section and explain the jump at the junction.
- A T-beam is loaded as shown - find the maximum tensile and compressive bending stresses.

Quick revision - Unit III

$$M/I = \sigma/y = E/R \qquad Z = I/y_{max} \qquad M = \sigma Z \qquad \tau = \frac{V A \bar{y}}{I b}$$

Section	τ_{max} / τ_{avg}	Location
Rectangle	1.5	N.A.
Circle	4/3	N.A.
Triangle	1.5	h/2 from apex
I-section	≈ 1.1 - 1.2	N.A.

- Important: derive $M/I = \sigma/y = E/R$ with assumptions.
- Important: derive $\tau = V A \bar{y} / (I b)$ and plot shear distribution for rectangular, I and T sections.
- Important: design problems; bending and shear stress problems on I and T sections.

UNIT - IV

DEFLECTION OF BEAMS & TORSION

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- DEFLECTION: bending into a circular arc, slope, deflection, radius of curvature
- Differential equation of the elastic line, double integration and Macaulay's methods
- Slope and deflection of cantilever and simply supported beams: point load, u.d.l., u.v.l.
- Mohr's theorems, moment-area method
- TORSION: derivation, torsion of circular shafts, pure shear
- Power transmission by circular shafts, shafts in series and parallel

Tip: memorise the standard slope and deflection table - it saves a lot of time in exams.

PART A - DEFLECTION OF BEAMS

1. Introduction

Under load a beam bends. The deflected shape of its neutral axis is the elastic curve. Limits on deflection are needed for serviceability (cracking of plaster, machine alignment). A common limit is $\delta_{max} \le L / 325$ for floors.

- Deflection y: vertical displacement of a point on the N.A. from its original position.

- Slope θ: angle between the tangent to the elastic curve and the original axis; $\theta \approx dy/dx$ for small deflections.
- Flexural rigidity EI: resistance of the beam to bending.

2. Bending into a circular arc

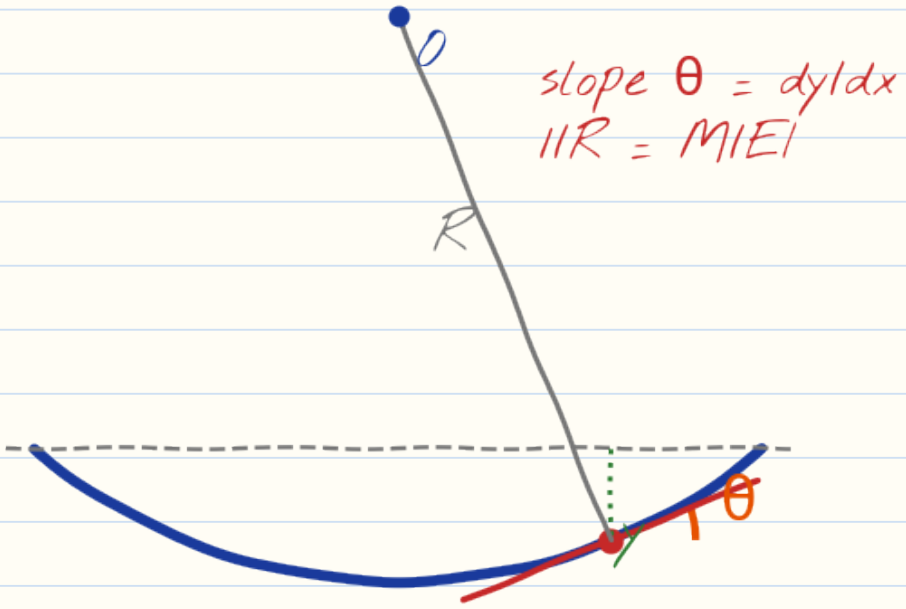


Fig: elastic curve, slope and radius of curvature R

From simple bending $M/I = E/R$, so the curvature is:

$$1/R = M/EI$$

From geometry the exact curvature of the curve $y(x)$ is $1/R = y''/[1 + (y')^2]^{3/2}$. For small slopes $(y')^2$ is negligible, so $1/R \approx d^2y/dx^2$.

3. Differential equation of the elastic line

$$EI\, d^2y/dx^2 = M(x)$$

Sign: with y measured upward positive and x to the right, a sagging (+ve) moment gives a curve concave upward, $y'' > 0$. Downward deflection therefore comes out negative - we quote the magnitude.

- $EI\, dy/dx = \int M\, dx + C_1$ (slope equation)

- $EIy = \iint M\,dx\,dx + C_1 x + C_2$ (deflection equation)
- C_1, C_2 from boundary conditions.

Support	Boundary condition
Fixed end	$y = 0$ and $\theta = 0$
Simple support (pin or roller)	$y = 0$
Free end	$M = 0$, $V = 0$
Symmetric beam, centre	$\theta = 0$

4. Double integration method

- 1. Find reactions and write $M(x)$ for the section at distance x (one expression per portion).
- 2. Put $M(x)$ in $EIy'' = M$ and integrate twice.
- 3. Apply boundary / continuity conditions to get C_1, C_2.
- 4. Substitute x to get slope or deflection; maximum deflection is where slope = 0.

Example 1: cantilever, point load W at free end

Take x from the fixed end: $M = -W(L - x)$.

$EIy'' = -W(L - x)$

$EIy' = -W(Lx - x^2/2) + C_1$; $y' = 0$ at $x = 0 \Rightarrow C_1 = 0$

$EIy = -W(Lx^2/2 - x^3/6) + C_2$; $y = 0$ at $x = 0 \Rightarrow C_2 = 0$

At $x = L$: $\theta_B = -WL^2/2EI$, $y_B = -WL^3/3EI$ (downward)

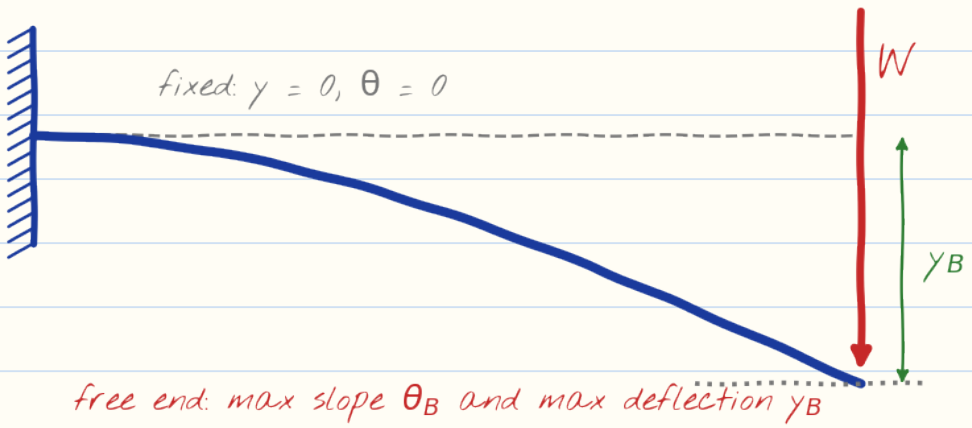


Fig: cantilever - maximum slope and deflection at the free end

Example 2: simply supported beam with u.d.l.

$M = w L x/2 - w x^2/2$

$E I y' = w L x^2/4 - w x^3/6 + C_1$

$E I y = w L x^3/12 - w x^4/24 + C_1 x + C_2$

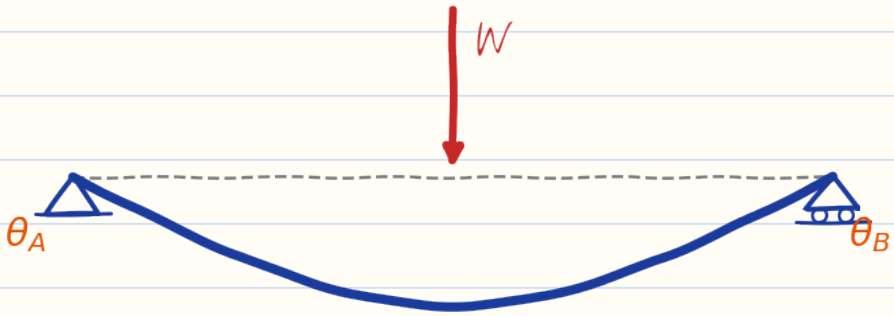
$y = 0$ at $x = 0 \Rightarrow C_2 = 0$; $y = 0$ at $x = L \Rightarrow C_1 = - w L^3/24$

Slope at A ($x = 0$): $\theta_A = - w L^3 / 24EI$

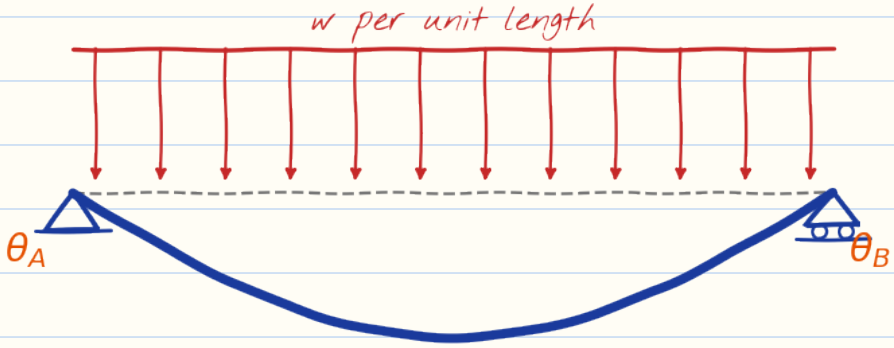
At centre $x = L/2$: $E I y = wL^4 [1/96 - 1/384 - 1/48] = - 5 w L^4/384$

$$y_{max} = 5 w L^4 / 384 E I$$

5. Standard results



$y_{max} = WL^3 / 48EI$ $\theta_A = \theta_B = WL^2 / 16EI$



$y_{max} = 5wL^4 / 384EI$ $\theta_A = \theta_B = wL^3 / 24EI$

Fig: S.S. beam - central point load and u.d.l.

Beam and loading	Slope (max)	Deflection (max)
Cantilever, W at free end	$WL^2 / 2EI$	$WL^3 / 3EI$
Cantilever, u.d.l. w	$wL^3 / 6EI$	$wL^4 / 8EI$
Cantilever, u.v.l. (0 at free end)	$wL^3 / 24EI$	$wL^4 / 30EI$
Cantilever, couple M at free end	ML / EI	$ML^2 / 2EI$
S.S., central W	$WL^2 / 16EI$	$WL^3 / 48EI$
S.S., u.d.l. w	$wL^3 / 24EI$	$5wL^4 / 384EI$
S.S., W at a from A, b from B	$\theta_A = Wb(L^2-b^2) / 6LEI$	under load: $Wa^2b^2 / 3EIL$

For a S.S. beam with u.v.l. (0 at A, w_0 at B): $\theta_A = 7w_0L^3/360EI$, $\theta_B = w_0L^3/45EI$, $y_{max} = 0.00652\, w_0L^4/EI$ at $x = 0.519L$.

6. Macaulay's method

Double integration needs a new pair of constants for every portion of a beam with several loads. Macaulay's method writes a single expression for M valid over the whole beam, using brackets $[x - a]$ that vanish when $x < a$. Then only two constants appear.

- Take the origin at the left end.
- Write M from the left, always on the section at the last portion (x in the right-most part).
- Write a term for each load: $W[x - a]$, $w[x - a]^2/2$, couple $M_0[x - a]^0$.
- A u.d.l. that stops before the end is extended to the end and an equal opposite u.d.l. is added.
- Integrate the brackets as a whole: $\int [x - a]\, dx = [x - a]^2/2$ (do NOT expand).
- A bracket is ignored if its value is negative ($x < a$).

Example 3: S.S. beam, point load W at a from A (b = L - a)

$R_A = W b / L$

$E I y'' = R_A x - W [x - a]$

$E I y' = R_A x^2/2 - W [x - a]^2/2 + C_1$

$E I y = R_A x^3/6 - W [x - a]^3/6 + C_1 x + C_2$

$x = 0, y = 0 \Rightarrow C_2 = 0.$ $x = L, y = 0 \Rightarrow C_1 = - W b (L^2 - b^2) / 6L$

So $\theta_A = - C_1/EI = W b (L^2 - b^2) / 6 L E I$ and under the load $y = W a^2 b^2 / 3 E I L$

Example 4: numerical

S.S. beam, span 5 m, point load 10 KN at 2 m from A. $E I = 1 \times 10^4$ KN·m². Find deflection under the load and slope at A.

$a = 2, b = 3, L = 5.$ $y = W a^2 b^2 / 3 E I L = 10 \times 4 \times 9 / (3 \times 1 \times 10^4 \times 5) = 2.4 \times 10^{-3}$ m = 2.4 mm

$\theta_A = W b (L^2 - b^2) / 6 L E I = 10 \times 3 \times 16 / (6 \times 5 \times 1 \times 10^4) = 1.6 \times 10^{-3}$ rad

7. Moment-area method (Mohr's theorems)

Draw the M/EI diagram. Between two points A and B of the elastic curve:

- Mohr's theorem I: the change of slope between A and B = area of the M/EI diagram between A and B. $\theta_{AB} = A_{AB}$
- Mohr's theorem II: the deviation of B from the tangent at A (measured perpendicular to the beam) = moment of the M/EI area between A and B about B. $\Delta_B = A_{AB} \cdot \bar{x}_B$

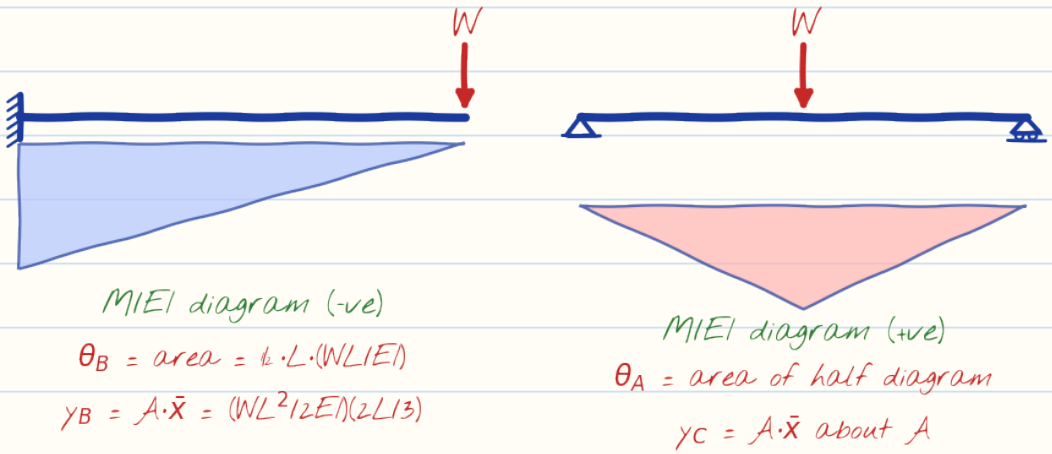


Fig: moment-area applications

Example 5: cantilever, W at free end

M/EI is a triangle: height W L / EI at the fixed end, area = ½ · L · W L/EI = W L²/2EI = θ_B

Centroid is 2L/3 from the free end: y_B = (W L²/2EI)(2L/3) = W L³/3EI

Example 6: S.S. beam, central W

Half of the M/EI diagram: area = ½ (L/2)(W L/4EI) = W L²/16EI = θ_A (tangent at C is horizontal)

y_C = moment of this area about A = (W L²/16EI)(L/3) = W L³/48EI

PART B - TORSION

8. Introduction

Torsion is the twisting of a member under a torque (twisting moment) T whose vector is along the axis - e.g. shafts of motors, turbines, drive shafts.

- Assumptions: the shaft is straight, circular and of uniform section; material homogeneous and elastic;
- plane cross-sections stay plane and radii stay straight after twisting;

- twist is uniform along the length, shear stress is proportional to shear strain ($\tau = G\gamma$).

9. Derivation of the torsion equation

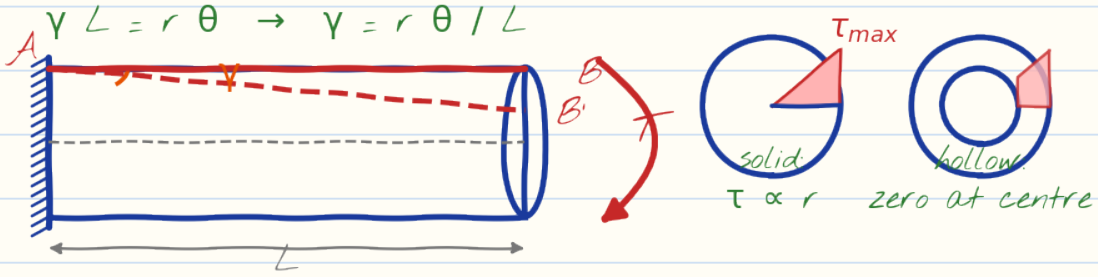


Fig: twisting of a shaft, shear stress on solid and hollow sections

A line AB on the surface moves to AB' – shear strain γ. For radius r and angle of twist θ over length L:

$\gamma L = r\theta \Rightarrow \gamma = r\theta / L$

$\tau = G\gamma = G r\theta / L \Rightarrow \tau / r = G\theta / L$ ...(1)

Moment of the force $\tau\, dA$ about the axis: $T = \int \tau r\, dA = (G\theta / L)\int r^2\, dA = G\theta J / L$ ...(2)

$$T / J = \tau / r = G\theta / L$$

- J = polar M.I. of the section (mm^4), G = modulus of rigidity, θ in radians.
- Solid shaft: $J = \pi d^4/32$; hollow: $J = \pi (D^4 - d^4)/32$.
- Torsional rigidity = GJ; polar modulus $Z_p = J / R$ (maximum shear stress $\tau_{max} = T / Z_p$).

Solid: $T = \pi \tau d^3 / 16$ Hollow: $T = \pi \tau (D^4 - d^4) / 16D$

10. Pure shear

A shaft element is in a state of **pure shear**: shear stress τ on the faces with no normal stress. Using the Mohr's circle (centre at the origin, radius τ): the principal stresses are $\sigma_1 = +\tau$ and $\sigma_2 = -\tau$ on planes at 45°.

- Ductile materials (mild steel) fail in shear on a plane perpendicular to the axis.
- Brittle materials (cast iron, chalk) fail along a 45° helix - on the plane of maximum tension.
- The relation $E = 2G(1+\mu)$ follows from this state.

11. Power transmitted by a shaft

$$P = T\omega = 2\pi N T / 60 \text{ (watts)} \qquad T = 60 P / (2\pi N)$$

N in rpm, T in N·m, P in W. (In KW: $T = 9550\, P / N$ N·m.)

Example 7

A solid shaft transmits 100 KW at 200 rpm. $\tau_{allow} = 60$ MPa, $G = 80$ GPa, $L = 2$ m. Find d and the angle of twist.

$T = 100\times10^3 \times 60 / (2\pi \times 200) = 4774.6$ N·m $= 4.7746\times10^6$ N·mm

$d^3 = 16T / (\pi\tau) = 16 \times 4.7746\times10^6 / (\pi \times 60) = 4.053\times10^5 \Rightarrow d = 74$ mm $\Rightarrow$ use 75 mm

$J = \pi \times 75^4 / 32 = 3.106\times10^6$ mm^4

$\theta = TL / GJ = 4.7746\times10^6 \times 2000 / (8\times10^4 \times 3.106\times10^6) = 0.0384$ rad $= 2.2°$

Example 8: hollow shaft

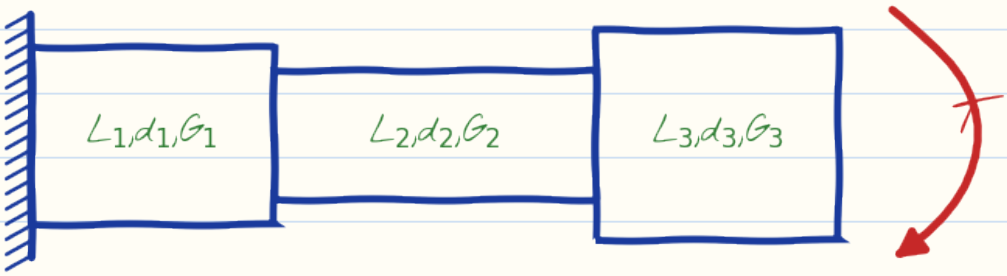
A hollow shaft $D = 100$ mm, $d = 80$ mm transmits T. If $\tau_{allow} = 50$ MPa find T.

$J = \pi(100^4 - 80^4)/32 = 5.797\times10^6$ mm^4, $Z_p = J/50 = 1.159\times10^5$ mm^3

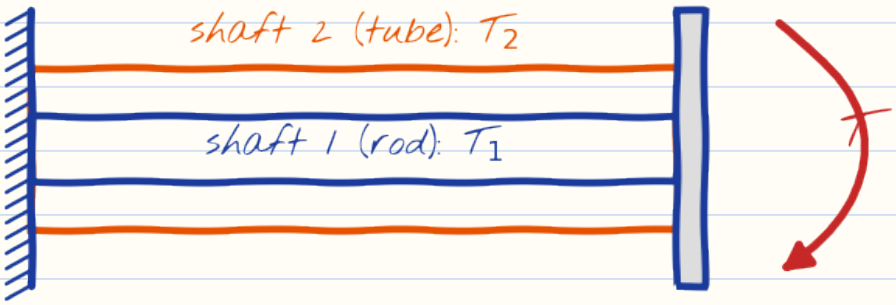
$T = \tau Z_p = 50 \times 1.159\times10^5 = 5.797\times10^6$ N·mm $= 5.80$ KN·m

A hollow shaft is more economical than a solid one: material near the axis carries little stress.

12. Shafts in series and in parallel



SERIES: same T in each, $\theta = \Sigma\, T L / G J$



PARALLEL: same θ, $T = T_1 + T_2$

- Series: shafts of different sections joined end to end carry the same torque. Total twist $\theta = \Sigma\, T L / (G J) = T L_1/G_1J_1 + T L_2/G_2J_2 + \dots$
- Parallel: shafts joined at both ends (e.g. a tube over a rod) have the same angle of twist, the torque is shared: $T = T_1 + T_2$.

$$T_1 / T_2 = (G_1J_1 / L_1) / (G_2J_2 / L_2) \qquad T_1 = T\, G_1J_1 / (G_1J_1 + G_2J_2) \text{ (same } L)$$

Example 9: shafts in parallel

A steel rod ($J_1 = 2\times10^6$ mm^4, $G = 80$ GPa) is inside a bronze tube ($J_2 = 6\times10^6$ mm^4, $G = 40$ GPa), same length, joined at ends. $T = 4$ KN·m. Find torque shares.

$G_1J_1 = 1.6\times10^{11}$, $G_2J_2 = 2.4\times10^{11}$

$T_1 = 4 \times 1.6/4.0 = 1.6$ KN·m, $T_2 = 2.4$ KN·m

More solved problems

Example 10: cantilever with u.d.l.

Cantilever $L = 2$ m, u.d.l. 10 KN/m, $E = 200$ GPa, $I = 8\times10^7$ mm^4. Find free-end slope and deflection.

$w = 10$ N/mm, $L = 2000$ mm, $EI = 2\times10^5 \times 8\times10^7 = 1.6\times10^{13}$ N·mm^2

$y_B = wL^4 / 8EI = 10 \times 1.6\times10^{13} / (8 \times 1.6\times10^{13}) = 1.25$ mm

$\theta_B = wL^3 / 6EI = 10 \times 8\times10^9 / (6 \times 1.6\times10^{13}) = 8.33\times10^{-4}$ rad

Example 11: S.S. beam, check against L/325

S.S. beam span 6 m, u.d.l. 12 KN/m, $E = 200$ GPa, $I = 1.5\times10^8$ mm^4.

$y_{max} = 5wL^4 / 384EI = 5 \times 12 \times (6000)^4 / (384 \times 2\times10^5 \times 1.5\times10^8) = 6.75$ mm

Permissible $= L/325 = 18.5$ mm $\Rightarrow$ safe in deflection ✓

Example 12: shaft - stress and twist

Solid shaft $d = 50$ mm, $L = 1.5$ m, $T = 1$ KN·m, $G = 80$ GPa.

$\tau = 16T / \pi d^3 = 16\times10^6 / (\pi \times 125\,000) = 40.7$ MPa

$J = \pi \times 50^4 / 32 = 6.136\times10^5$ mm^4

$\theta = TL / GJ = 10^6 \times 1500 / (8\times10^4 \times 6.136\times10^5) = 0.0306$ rad $= 1.75°$

Example 13: stepped shaft (series)

Shaft AB: $d = 60$ mm, $L = 1$ m; BC: $d = 40$ mm, $L = 1$ m; $T = 500$ N·m; $G = 80$ GPa.

$J_1 = 1.272\times10^6$, $J_2 = 2.513\times10^5$ mm^4

$\theta = (T L / G)(1/J_1 + 1/J_2) = 6.25 \times 4.765 \times 10^{-6} = 0.0298$ rad = 1.71°

Maximum shear stress is in the thinner part BC: $\tau = 16 \times 5 \times 10^5 / (\pi \times 40^3) = 39.8$ MPa

Strength and stiffness criteria

- A shaft is designed for strength ($\tau \leq \tau_{allow}$) and checked for stiffness (θ per metre $\leq$ limit, often 0.25° to 1° per metre).
- Take the larger diameter obtained from the two criteria.
- Beams: design for bending (strength), then verify the deflection limit L/325 or as specified.
- Deflection varies with L^3 or L^4 - doubling the span makes the beam far more flexible; doubling depth reduces deflection to 1/8.

Important questions

- Derive the differential equation of the elastic curve: $E I \, d^2y/dx^2 = M$.
- Find slope and deflection of a cantilever with u.d.l. using double integration.
- Explain Macaulay's method; find the deflection under the load of a S.S. beam.
- State and prove Mohr's theorems; use them to find the deflection of a cantilever.
- Derive $T/J = \tau/r = G\theta/L$ with assumptions.
- Compare solid and hollow shafts. A shaft transmits 150 KW at 300 rpm - find the diameter.

Quick revision - Unit IV

$EI\,y'' = M$ $\quad 1/R = M/EI$ $\quad \theta = \int M/EI\,dx$ $\quad y_{max}$ (S.S., udl) $= 5wL^4/384EI$

$T/J = \tau/r = G\theta/L$ $\quad P = 2\pi NT/60$ $\quad$ series: same T $\quad$ parallel: same θ

- Important: derive $y = WL^3/3EI$ (cantilever) and $5wL^4/384EI$ (S.S. udl) by double integration.
- Important: Macaulay's method problems; moment-area theorems; torsion equation derivation; power transmission; shafts in series/parallel.

UNIT - V

THIN & THICK CYLINDERS, COLUMNS

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- THIN CYLINDERS: seamless shells, longitudinal and circumferential stresses
- Hoop, longitudinal and volumetric strains; changes in diameter and volume
- Thin spherical shells; wire-wound thin cylinders
- THICK CYLINDERS: Lamé's equations, inside and outside pressures, compound cylinders
- COLUMNS: buckling and stability, Euler's formula for various end conditions
- Limitations of Euler's formula, Rankine's formula

Tip: thin cylinder: $\sigma_c = pd/2t$ and $\sigma_L = pd/4t$. Column: $P_{cr} = \pi^2 EI/L_e^2$.

PART A - THIN CYLINDERS AND SHELLS

1. Thin cylinders

A cylindrical shell is thin if the wall thickness t is small compared with the internal diameter d: $t < d/20$. Then the stress is assumed uniform across the thickness and the radial stress ($\approx p$ at the inside, 0 outside) is ignored.

Internal pressure p causes two principal stresses in the wall:

- Circumferential (hoop) stress σ_c - tries to split the cylinder along a length.

- Longitudinal stress σ_L - tries to tear the cylinder across a section.

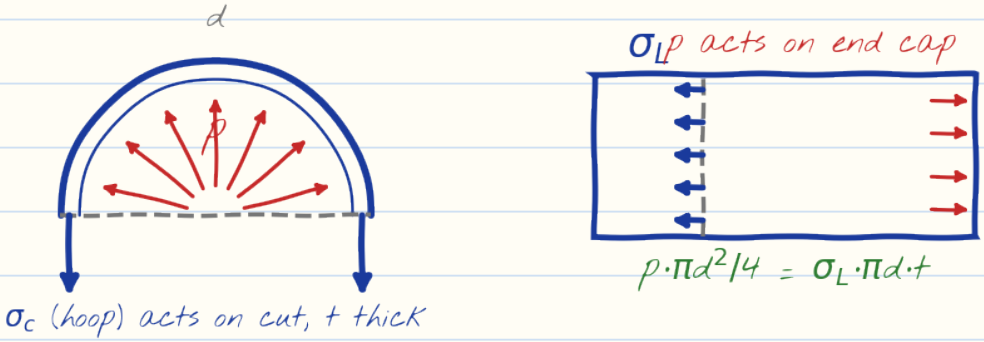


Fig: forces on a thin cylinder - hoop stress (left) and longitudinal stress (right)

2. Derivation of the stresses

Circumferential stress: cut the cylinder along a diameter, length L. Bursting force = p x d x L; resisting force = 2 σ_c t L.

$p\,d\,L = 2\,\sigma_c\,t\,L \Rightarrow \sigma_c = p\,d\,/\,2t$

Longitudinal stress: cut across. Bursting force = p x $\pi\,d^2/4$; resisting = σ_L x $\pi\,d\,t$.

$p\,\pi\,d^2/4 = \sigma_L\,\pi\,d\,t \Rightarrow \sigma_L = p\,d\,/\,4t$

$$\sigma_c = p\,d\,/\,2\,t \qquad \sigma_L = p\,d\,/\,4\,t = \sigma_c\,/\,2$$

Hoop stress is twice the longitudinal stress - so longitudinal joints must be stronger than circumferential joints.

Maximum shear stress (in the plane): $\tau_{max} = (\sigma_c - \sigma_L)/2 = p\,d\,/\,8t$. Including the radial stress it is $p\,d\,/\,4t$.

3. Efficiency of joints

If the shell has riveted/welded joints, the plate is weakened:

$\sigma_c = p\,d\,/\,(2 + \eta_l)$ (η_l = efficiency of longitudinal joint)

$\sigma_L = p\,d\,/\,(4 + \eta_c)$ (η_c = efficiency of circumferential joint)

4. Strains and volume change

Using σ_c and σ_l as biaxial stresses (Poisson effect):

$$\varepsilon_c = (\sigma_c - \mu\,\sigma_l)/E = (p\,d/4t)(2 - \mu)$$
$$\varepsilon_l = (\sigma_l - \mu\,\sigma_c)/E = (p\,d/4t)(1 - 2\mu)$$
$$\varepsilon_v = 2\,\varepsilon_c + \varepsilon_l = (p\,d/4t)(5 - 4\mu)$$

- Circumferential strain $\varepsilon_c = \delta d / d$ ⇒ change in diameter $\delta d = \varepsilon_c\, d$
- Longitudinal strain $\varepsilon_l = \delta L / L$ ⇒ change in length $\delta L = \varepsilon_l\, L$
- Volumetric strain $\varepsilon_v = \delta V / V \Rightarrow \delta V = \varepsilon_v\, V$ ($V = \pi d^2 L / 4$)

Example 1

A thin cylinder $d = 800$ mm, $t = 10$ mm, $L = 2$ m carries $p = 2$ MPa. $E = 2\times10^5$ MPa, $\mu = 0.3$. Find stresses, strains and changes in d and V.

$\sigma_c = 2 \times 800 / (2 \times 10) = 80$ MPa, $\sigma_l = 40$ MPa

$\varepsilon_c = (80 - 0.3\times40)/2\times10^5 = 3.4\times10^{-4}$; $\varepsilon_l = (40 - 24)/2\times10^5 = 8\times10^{-5}$

$\delta d = 3.4\times10^{-4} \times 800 = 0.272$ mm, $\delta L = 8\times10^{-5} \times 2000 = 0.16$ mm

$\varepsilon_v = 2(3.4\times10^{-4}) + 0.8\times10^{-4} = 7.6\times10^{-4}$

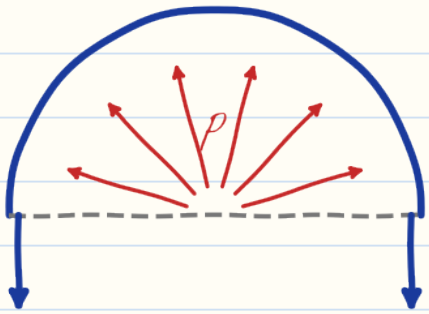
$V = \pi/4 \times 800^2 \times 2000 = 1.0053\times10^9$ mm^3 ⇒ $\delta V = 7.64\times10^5$ mm^3

Example 2: thickness

Find the thickness of a thin cylinder $d = 600$ mm for $p = 3$ MPa if $\sigma_{allow} = 100$ MPa.

$t = p\,d / 2\,\sigma = 3 \times 600 / (2 \times 100) = 9$ mm (use 10 mm)

5. Thin spherical shells



$\sigma = pd / 4t$

$p \pi d^2/4 = \sigma \pi d t \Rightarrow \sigma = p d / 4t$ (same in every direction)

$\varepsilon = (\sigma/E)(1 - \mu) = p d (1 - \mu) / 4 t E$ $\quad \varepsilon_v = 3 \varepsilon = 3 p d (1 - \mu) / 4 t E$

- $\delta d = \varepsilon d$, $\delta V = 3 \varepsilon V$ $(V = \pi d^3/6)$.
- For the same d, t and p the spherical shell has half the stress of the cylinder (hoop) - sphere is the most efficient pressure vessel shape.
- Cylinder with hemispherical ends: equal change of diameter requires $t_s/t_c = (1 - \mu)/(2 - \mu)$.

6. Wire-wound thin cylinders

A thin cylinder is wound with a layer of steel wire under initial tension. The wire puts the cylinder in initial circumferential compression, so that under working pressure the resultant stress is smaller - the cylinder resists higher pressure.

- Let wire stress σ_{wi} (tension) and cylinder stress σ_{ci} (compression) after winding, thicknesses t_w and t_c:
- Equilibrium after winding: $\sigma_{ci} t_c = \sigma_{wi} t_w$.
- Under pressure p the extra stresses are in the ratio of their moduli: $\Delta\sigma_w / E_w = \Delta\sigma_c / E_c$.
- Equilibrium of pressure: $\Delta\sigma_c t_c + \Delta\sigma_w t_w = p d / 2$.

$$\Delta\sigma_c = (p\,d/2)/(t_c + (E_w/E_c)\,t_w)$$

Final cylinder stress $= \Delta\sigma_c - \sigma_{ci}$ Final wire stress $= \sigma_{wi} + \Delta\sigma_w$

PART B - THICK CYLINDERS

7. Lamé's theory

When $t > d/20$ the hoop stress varies markedly across the wall, so it cannot be taken uniform. Lamé derived the stresses at radius r for a thick cylinder (internal radius r_1, external r_2).

- Assumptions: material homogeneous, isotropic, elastic; plane sections remain plane (longitudinal strain uniform); stresses are axisymmetric.
- Radial pressure p_r and hoop stress σ_θ at radius r:

$$p_r = B/r^2 - A \qquad \sigma_\theta = B/r^2 + A$$

(Lamé's equations)

A and B are constants found from the boundary conditions: $p_r = p_i$ at $r = r_1$, and $p_r = p_o$ at $r = r_2$:

$$A = (p_i r_1^2 - p_o r_2^2)/(r_2^2 - r_1^2) \qquad B = (p_i - p_o)\, r_1^2 r_2^2/(r_2^2 - r_1^2)$$

- Longitudinal stress (closed ends): $\sigma_L = A = (p_i r_1^2 - p_o r_2^2)/(r_2^2 - r_1^2)$.
- Radial stress is compressive and equals $-p_r$.
- Maximum shear stress $= (\sigma_\theta + p_r)/2$.

8. Internal pressure only ($p_o = 0$)

$$\sigma_\theta = (p_i r_1^2/(r_2^2 - r_1^2))(1 + r_2^2/r^2)$$
$$p_r = (p_i r_1^2/(r_2^2 - r_1^2))(r_2^2/r^2 - 1)$$

$\sigma_{\theta,max}$ (at r_1) $= p_i (r_2^2 + r_1^2)/(r_2^2 - r_1^2)$ σ_θ (at r_2) $= 2 p_i r_1^2/(r_2^2 - r_1^2)$

Example 3

Thick cylinder: internal dia 160 mm, external dia 240 mm, internal pressure 20 MPa. Find the stresses.

$r_1 = 80$, $r_2 = 120$. $A = 20 \times 6400 / (14400 - 6400) = 16$; $B = 16 \times 14400 = 230\,400$

σ_θ at inner $= 230400/6400 + 16 = 52$ MPa; at outer $= 230400/14400 + 16 = 32$ MPa

radial: at inner $= -20$ MPa (p_i); at outer $= 0$. $\sigma_L = A = 16$ MPa

τ_{max} (inner) $= (52 + 20)/2 = 36$ MPa

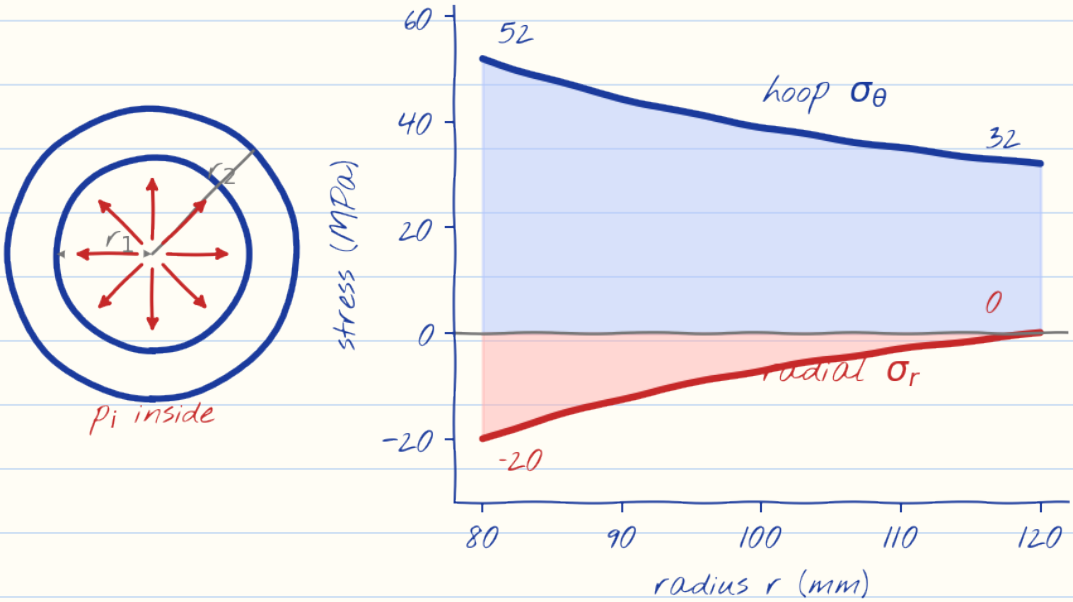


Fig: variation of hoop and radial stress across the wall (Example 3)

9. Design of thick cylinder

For a permissible hoop stress σ_t equate the maximum hoop stress at the inside:

$$r_2 = r_1 \sqrt{[(\sigma_t + p_i) / (\sigma_t - p_i)]} \qquad \text{thickness } t = r_2 - r_1$$

Note: if $p_i \geq \sigma_t$, no thickness is enough - thickening only helps up to a limit. Use a compound cylinder.

Example 4

$r_1 = 100$ mm, $p_i = 40$ MPa, $\sigma_t = 100$ MPa:

$r_2 = 100\sqrt{(140/60)} = 152.8$ mm $\Rightarrow$ $t = 52.8$ mm

10. External pressure and compound cylinders

- With both p_i and p_o use the general A and B above.
- Only external pressure p_o ($p_i = 0$): the maximum hoop stress is at the inner surface and is compressive: $\sigma_\theta = -2p_o r_2^2 / (r_2^2 - r_1^2)$.

Compound cylinder: an outer cylinder with a slightly smaller bore is heated and shrunk on to an inner cylinder (or pressed on). After cooling a shrink-fit pressure p_s acts at the junction.

- The inner cylinder is put in hoop compression, the outer in hoop tension.
- Under working pressure the stresses add to this initial state (superposition) - hoop stress across the wall becomes nearly uniform.
- The shrinkage allowance (difference of radii) = hoop strain difference at the junction x radius.
- Procedure: (i) hoop stress due to p_s in each cylinder with Lamé; (ii) stresses due to p_i on the combined cylinder; (iii) add algebraically.
- Benefit: lower peak stress, so the compound cylinder is stronger than a single one of the same material and thickness.

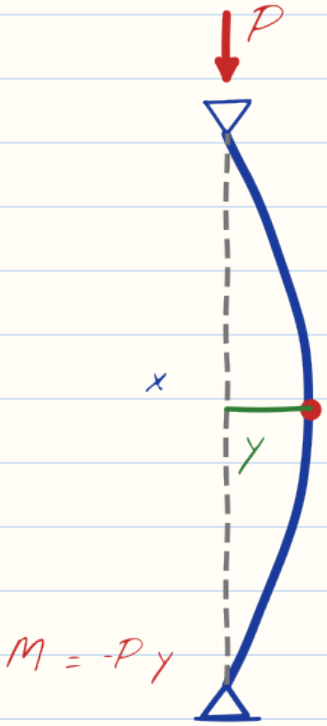
PART C - COLUMNS

11. Buckling and stability

A column (strut) is a compression member whose length is large compared with its lateral dimensions. A short column fails by crushing; a long slender column fails by buckling - sudden lateral bending at a load below the crushing load.

- Stable equilibrium: a small side push is recovered when removed.
- Unstable: the displacement keeps growing.
- Critical (buckling) load P_{cr}: the load at which the straight column is in neutral equilibrium – the axial load that can hold the column slightly bent.

12. Euler's formula – both ends pinned



Assumptions: column initially straight, axially loaded, material elastic and homogeneous, Euler's theory ignores the effect of axial shortening, and the failure is by buckling alone.

Bending moment at a section at distance x: $M = -Py$

$EI\, d^2y/dx^2 = -Py \Rightarrow d^2y/dx^2 + (P/EI)\, y = 0$

Solution: $y = C_1 \cos(\alpha x) + C_2 \sin(\alpha x)$, $\alpha^2 = P/EI$

$y = 0$ at $x = 0 \Rightarrow C_1 = 0$; $y = 0$ at $x = L \Rightarrow C_2 \sin(\alpha L) = 0 \Rightarrow \alpha L = n\pi$

Smallest load: $n = 1 \Rightarrow \alpha = \pi/L$

$$P_{cr} = \pi^2 E I / L^2$$

Buckling occurs about the axis of least moment of inertia, so use I_{min}.

13. Other end conditions – effective length

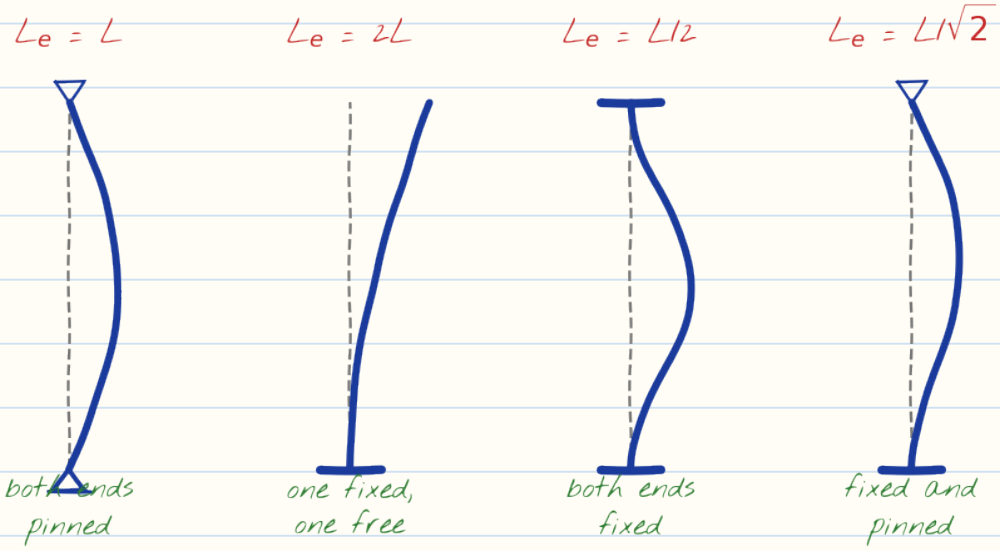


Fig: buckled shapes and effective lengths

$$P_{cr} = \pi^2 E I / L_e^2$$

End conditions	L_e	P_{cr}	Ratio
Both ends pinned	L	$\pi^2 EI/L^2$	1
One fixed, one free	$2L$	$\pi^2 EI/4L^2$	1/4
Both fixed	$L/2$	$4\pi^2 EI/L^2$	4
One fixed, one pinned	$L/\sqrt{2}$	$2\pi^2 EI/L^2$	2

14. Slenderness ratio and limitations of Euler's formula

Slenderness ratio $\lambda = L_e / r_{min}$ ($r = \sqrt{(I/A)}$ radius of gyration)

$$\sigma_{cr} = P_{cr}/A = \pi^2 E / \lambda^2$$

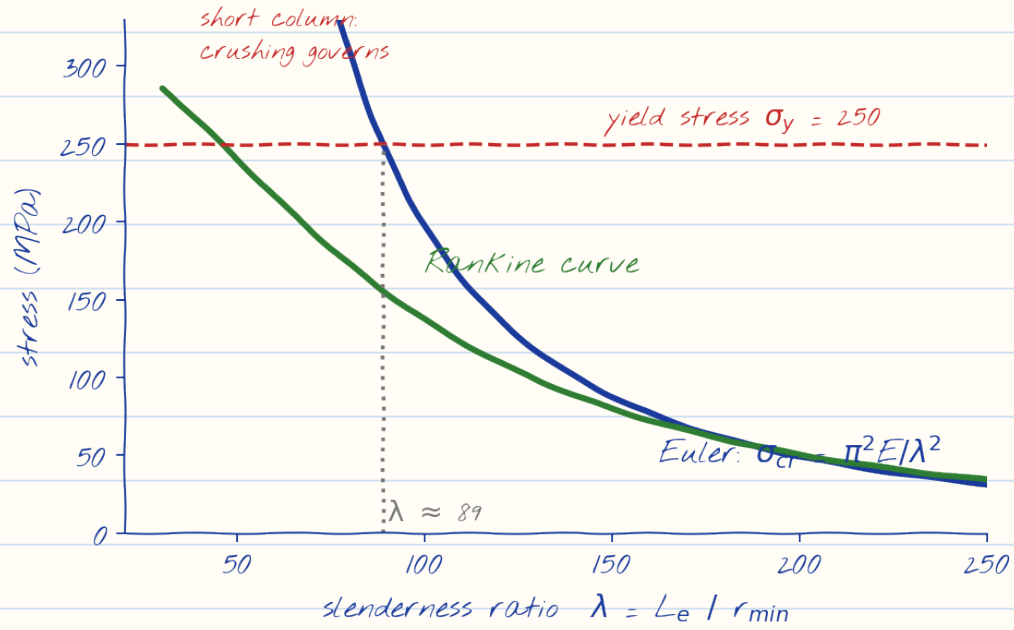


Fig: critical stress vs slenderness ratio (mild steel)

- Euler's formula is valid only while $\sigma_{cr} \leq$ the proportional limit (or yield stress): $\lambda \geq \lambda_{limit} = \sqrt{(\pi^2 E / \sigma_y)}$.
- For mild steel ($E = 2 \times 10^5$, $\sigma_y = 250$ MPa): $\lambda_{limit} \approx 89$ (often taken ≈ 80 to 100).
- For short columns (λ small) Euler gives a stress far above the yield stress - absurd; use Rankine or Johnson/Secant.
- The formula neglects initial crookedness, eccentricity of load and imperfect end conditions - so a factor of safety (about 3 for steel) is applied.

15. Rankine's formula

An empirical formula covering short as well as long columns. If $P_c = \sigma_c A$ is the crushing load and P_E the Euler load, then (Rankine-Gordon):

$$1/P_R = 1/P_c + 1/P_E \quad \Rightarrow \quad P_R = \sigma_c A / [1 + a (L_e/r)^2]$$

$$a = \sigma_c / (\pi^2 E) \text{ (Rankine constant)}$$

Material	σ_c (MPa)	a
Mild steel	320	1/7500
Cast iron	550	1/1600
Wrought iron	250	1/9000
Timber	50	1/750

- For small λ, $P_R \to \sigma_c A$ (crushing); for large λ, $P_R \to P_E$ (Euler).
- Rankine's formula is simple and widely used in design; safe load = P_R / FOS.

Example 5

A 3 m long solid steel column, 60 mm dia, both ends pinned. E = 200 GPa. Find P_{cr} (Euler) and the Rankine load (σ_c = 320 MPa, a = 1/7500).

$I = \pi \times 60^4 / 64 = 6.362 \times 10^5$ mm^4; $A = 2827.4$ mm^2; $r = d/4 = 15$ mm; $\lambda = 3000/15 = 200$

$P_{cr} = \pi^2 E I / L^2 = 9.8696 \times 2\times10^5 \times 6.362\times10^5 / 3000^2 = 139.5$ KN

$\sigma_{cr} = \pi^2 E/\lambda^2 = 49.3$ MPa ($< \sigma_y$ - Euler valid since $\lambda = 200 > 89$)

$P_R = 320 \times 2827.4 / (1 + 200^2/7500) = 904\,768 / 6.333 = 142.9$ KN

Example 6: end conditions

The same column with both ends fixed: $L_e = 1.5$ m, $\lambda = 100$.

$P_{cr} = 4 \times 139.5 = 558$ KN, $\sigma_{cr} = 197.4$ MPa (< 250, Euler still valid since $\lambda = 100 > 89$)

Example 7: hollow column

A hollow circular column D = 100 mm, d = 80 mm, L = 4 m, hinged at both ends, E = 200 GPa.

$I = \pi (100^4 - 80^4)/64 = 2.898\times10^6$ mm^4

$P_{cr} = \pi^2 \times 2\times10^5 \times 2.898\times10^6 / 4000^2 = 357.5$ KN

Safe load with FOS 3 = 357.5 / 3 = 119 KN

Quick revision - Unit V

$\sigma_c = pd/2t$ $\sigma_L = pd/4t$ sphere $\sigma = pd/4t$ $\varepsilon_v = (pd/4tE)(5 - 4\mu)$

Lame: $\sigma_\theta = B/r^2 + A$ $p_r = B/r^2 - A$ $P_{cr} = \pi^2 EI/L_e^2$ $P_R = \sigma_c A/(1 + a\lambda^2)$

- Important: derive σ_c and σ_L; change in volume of a thin cylinder; thin sphere.
- Important: Lame's equations and stress distribution; thickness of thick cylinder; compound cylinder.
- Important: Euler's formula for different end conditions; limitations; Rankine's formula problems.